Cycle Financing and Near-Convergent Diophantine Obstructions in the Juggler Map

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Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}. \end{cases}$$

It is conjectured that every positive integer eventually reaches 1. This paper does not prove that conjecture. It proves period lower bounds for a hypothetical nontrivial cycle, once a verified descent floor is given.

We develop a cycle-financing inequality for this floor-power map. Exact integer one-step preimages give a one-step logarithmic defect; cycle minimality lets that defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. For a hypothetical cycle of length L with o odd steps and minimum n,

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

This inequality is the main theorem; every numerical period bound below is an instantiation of it, or of its coupled walk-charge refinement, at a certified descent floor. As a consequence, combining the inequality with the known verification through 10^6 yields $L \geq 25781$; at the laboratory-certified descent floor $N_0 = 26254995$ the same table yields $L \geq 50508$. The novelty is not a new computational record; it is that implication. A walk-charge envelope — transport of the floor losses to a reduced base, identification of the extremal exponent walk as a rotation itinerary, and a Denjoy–Koksma bound over certified Ostrowski blocks, census-free on the window $[50508, 16785921)$ — then extends the exclusion at the laboratory floor: any nontrivial cycle has period at least 176251. The same certified kill criterion at a second certified floor, $N_0 = 162849448$, on the surviving lengths — all inside the census-free window, which reaches the end of the fan — gives period at least 478245. The main numerical result evaluates it once more at the third certified floor $N_0 = 350000000$: any nontrivial Juggler cycle has period at least 780239. We also prove that every nontrivial cycle contains at least four even steps, and hence has period at least eleven; that bound uses no descent floor. A floor-free gap transfer, $n \log n \cdot \min(o \log 3 - L \log 2,\, 1) \leq 2L$, combined with Rhin's effective measure, excludes every cycle with $L^{14.3} \leq n \log n / 915$ and reduces the no-cycle problem to the long regime, where it remains open. The core lemmas are formalized in Lean 4; the descent floors and the per-length kill tables are independently certified computations, and the paper does not claim to be formally verified as a whole. Two companion manuscripts use this paper's power envelope and certified floor as inputs: a parity-discrepancy paper (depth-4 equidistribution of nested floor powers, certificate density 7/8) and a fate-contagion paper, in which the floor is the target of a Tao-type reduction and the basin of any hypothetical cycle is shown to have logarithmic count $\gg (\log x)^{0.405}$; Section 6.1 records what they add to the cycle problem and what they do not.

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1. Introduction

The Juggler sequence was introduced by Pickover [1,2] as an interesting variation of the Collatz problem. Pickover's later exposition is Chapter 45 of [2], pp. 102–106. The object of study is the one-step map $J : \mathbb{N} \to \mathbb{N}$ displayed in the abstract. The *Juggler sequence* starting at n is the trajectory of iterates n, $J(n)$, $J^2(n), \ldots$. The On-Line Encyclopedia of Integer Sequences records the one-step values of J as A094683 [3] and the number of steps to reach 1 (when that occurs) as A007320 [4]; those catalogue entries are not theorems of this paper.

The trajectory of 3 is $3, 5, 11, 36, 6, 2, 1$. The trajectory of 37 already peaks at 24906114455136. Universal arrival at 1 remains an open conjecture. This paper does not prove that conjecture. It proves period lower bounds for a *hypothetical* nontrivial cycle, once a verified descent floor is given.

Throughout, $\mathbb{N} = \{1, 2, 3, \ldots\}$. Write J^k for the k-fold iterate, and write $\lfloor x \rfloor$ for the integer part of a real $x \geq 0$: discard the fractional part. Thus $\lfloor 5.196\ldots \rfloor = 5$ and $\lfloor 6 \rfloor = 6$. The floor is applied after every branch of J, not once at the end of a walk.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \qquad O(n) = \lfloor n^{3/2} \rfloor,$$

so $J(n) = E(n)$ when n is even and $J(n) = O(n)$ when n is odd.

Trajectory. The *trajectory* of $n \in \mathbb{N}$ is the sequence of values $n_0 = n$, $n_{i+1} = J(n_i)$ (also called an orbit in some dynamics texts).

Itinerary. The *itinerary* of the first k steps is the length-k string $w \in \{O, E\}^k$ with $w_i = O$ if n_i is odd and $w_i = E$ if n_i is even. Synonyms: itinerary = parity sequence; trajectory = orbit. The itinerary is the list of branch labels, not the list of values. The integer k is any finite prefix length; the definition does not assume that the trajectory reaches 1.

Ideal exponent. An itinerary of length k with o odd letters has ideal exponent $3^o/2^k$. Ignoring floors, those letters would multiply the start by that ratio. Floors make the actual image smaller.

Realized itinerary. An itinerary w is *realized* at n when the first $|w|$ parities of the trajectory of n are exactly the letters of w. Equivalently: a formal string over $\{O, E\}$ is an itinerary only when some start actually follows those branches. That is what is meant by: an itinerary is available only when the trajectory realizes those parities. Floors are applied after every letter of a realized itinerary.

One-step preimage. The map J is not invertible. For $m \in \mathbb{N}$, the *one-step preimage* of m is the set

$$J^{-1}(m) = \{k \in \mathbb{N} : J(k) = m\}.$$

An even image has a whole interval of even parents; an odd image has at most one odd parent (Lemma 3.1).

Lemma 1.1 (three fates). Let $n \in \mathbb{N}$. The trajectory of n does exactly one of the following: (i) some iterate equals 1; (ii) some iterate $m \geq 2$ returns, and the trajectory is eventually periodic through a cycle containing m; (iii) the trajectory is unbounded. In particular, a bounded infinite trajectory is eventually periodic.

Proof. The map J is a well-defined function $\mathbb{N} \to \mathbb{N}$, so the forward trajectory is an infinite sequence in $\mathbb{N}$. If some term equals 1, we are in (i); note $J(1) = 1$. If the trajectory is unbounded, we are in (iii). If the trajectory is infinite and bounded, the pigeonhole principle supplies a repeated value m; uniqueness of the successor $J(m)$ then forces a cycle, which is (ii) unless $m = 1$, already covered by (i). □

The lemma lists the only logical possibilities. It does not assert which fate occurs for a given start, and it is not a termination theorem.

Object	Meaning
J	the one-step map

Object	Meaning
E, O	even and odd branches
trajectory	the sequence of values
itinerary	a finite string in $\{O, E\}$ of parities
realized itinerary	the trajectory actually follows those letters
$3^o/2^k$	ideal exponent of an itinerary of length k with o odd letters
$J^{-1}(m)$	the one-step preimage of m
N_0	a verified descent floor (a computational input)
CycleMin	a minimum-based rotation of a cycle itinerary

The mechanism is the interaction of three elementary facts: exact integer one-step preimages, a logarithmic defect, and cycle minimality. Exact one-step preimages give a one-step logarithmic defect; cycle minimality lets the defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. That is Theorem 4.4:

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

The inequality $\log(1+u) \leq u$ is the only analytic input; it is not the content of the theorem.

Write N_0 for a *verified descent floor*: every start $2 \leq n \leq N_0$ reaches 1. A floor is an input, not the result. Four instances are used. The base instance is $N_0 = 10^6$, reported by Weisstein [5] and recomputed here by exact first-passage. Combined with Theorem 4.4 it gives

$$\text{known verification through } 10^6 \ + \ \text{this inequality} \quad \Rightarrow \quad L \geq 25781.$$

The laboratory instance is $N_0 = 26254995$, certified by the same exact first-passage method (Proposition 5.1); the same table then gives $L \geq 50508$ (Theorem 5.2), and the walk-charge envelope of Section 5 amplifies it to $L \geq 176251$ (Theorem 5.9). Corollary 5.10 evaluates the same kill criterion at the second certified floor $N_0 = 162849448$, every survivor lying inside the $[50508, 16785921)$ window of Theorem 5.8, whose envelope therefore covers their charge, and gives $L \geq 478245$. The main numerical result is Corollary 5.11: at the third certified floor $N_0 = 350000000$ the same comparison gives $L \geq 780239$. The census-free window theorem covers $[50508, 16785921)$ — the whole semiconvergent fan of Proposition 5.12 — so the charge side of the later-floor kills needs no per-length dynamic program. The kill comparison itself stays per-length, because its left-hand side $\theta(L)$ is a Diophantine quantity the envelope does not control. The architecture is

$$\text{envelope} \to \text{cycle minimum} \to \text{finance} \to n_{\max}(L) \to N_0 \to L \geq 25781 \text{ at } 10^6,$$

extended at the laboratory floor by

$$\text{transport} \to \text{hug adversary} \to \text{itinerary identity} \to \text{Denjoy–Koksma} \to \text{window} \to L \geq 176251.$$

The computation supplies the endpoint N_0. The mathematics amplifies that floor to the period bound.

Roadmap. Section 2 records the power envelope and the exact defect identity that explains it. Section 3 classifies minimum-based cycle itineraries and proves that every nontrivial cycle has at least four even letters. Section 4 records the excursion necklace of a minimum-based itinerary, unrolls the one-step-preimage logarithm around that minimum, obtains the finance inequality, and applies it at the known floor 10^6. The necklace is the geometry of the unroll, not a fourth main theorem. A run-type packing of the same identity is a supporting refinement; the arithmetic of the leftover lengths is secondary. Section 5 certifies the laboratory floor 26254995, replaces the length-only charge by a coupled exponent-walk charge, identifies its extremal word as a rotation itinerary, and proves a census-free envelope for every length in the window $[50508, 16785921)$; the resulting kill table gives the period bound 176251, a certified evaluation of the same kill criterion at a second certified floor raises it to 478245 (Corollary 5.10), and a third certified floor raises it to 780239 (Corollary 5.11). Section 6 records limitations.

A nonempty realized itinerary w with $J^{|w|}(n) = n$ is a *cycle itinerary*. The unique fixed point is 1; a cycle is *nontrivial* when it contains some $n \geq 2$. A cycle word at n is *minimum-based* when n is a cycle minimum: $J^j(n) \geq n$ for every $0 \leq j < |w|$. Every cycle itinerary has a minimum-based rotation.

All exceptional sets below are *finance-survivor* sets: lengths that a stated form of the finance inequality does not exclude at the verified descent floor. They are not candidate cycle sets. A survivor count "through X" is inclusive: it counts every surviving length $L \leq X$.

1.0 Main results

Contribution 1 — cycle-financing inequality. For a minimum-based cycle of length L with o odd steps,

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o$$

(Theorem 4.4). Floor defects become a quantitative bound on the cycle minimum.

Contribution 2 — structural itinerary obstruction. Every nontrivial cycle itinerary has at least four even letters, and hence period at least eleven (Theorem 3.22 and Corollary 3.23). The argument classifies the minimum-based itinerary geometry; it is not a raw census of itineraries of length at most ten. This bound does not use the verified descent floor.

Contribution 3 — explicit conditional consequence for hypothetical cycles. Combined with the independently verified descent floor $N_0 = 10^6$, there is no nontrivial Juggler cycle of length at most 25780. Equivalently, any nontrivial cycle has period at least 25781 (Theorem 4.6(A)). At the laboratory-certified floor $N_0 = 26254995$ the same table gives period at least 50508 (Theorem 5.2).

Contribution 4 — walk-charge envelope and the main period bound. On a minimum-based cycle every state is coupled through one closed exponent walk. Transport of the floor losses to a reduced base (Theorem 5.3), identification of the extremal walk as the rotation (hug) itinerary (Theorem 5.4, Lemma 5.6), and a Denjoy–Koksma bound over certified Ostrowski blocks (Theorem 5.7) give a uniform envelope for every length in the window $[50508, 16785921)$ (Theorem 5.8) — census-free on that window, which covers the whole semiconvergent fan. At the laboratory floor the resulting kill table leaves a single finance survivor below $2 \cdot 10^5$: any nontrivial cycle has period at least 176251 (Theorem 5.9), the first laboratory instance. The second-floor evaluation is Corollary 5.10: at $N_0 = 162849448$ the same kill criterion, evaluated on the additional survivors — all inside the window, so all with a census-free charge bound — leaves only the semiconvergent fan member 478245. The main numerical result is Corollary 5.11: at the third certified floor $N_0 = 350000000$ that comparison leaves only the next fan member 780239: any nontrivial cycle has period at least 780239.

Contribution 5 — floor-free gap transfer and the short-cycle reduction. The finance inequality ties the relative surplus to the linear form $\Lambda = o \log 3 - L \log 2$ without any floor: $n \log n \cdot \min(\Lambda, 1) \leq 2L$ (Theorem 4.10, Lean `cycleMin_gap_transfer`). With Rhin's effective measure this excludes every cycle with $L^{14.3} \leq n \log n / 915$ (Corollary 4.11), for all $n \geq 2$. The statement is a reduction, not a kill: it is weaker than the finance table at every certified floor, and it shows that the no-cycle problem is exactly the exclusion of long cycles, $L > (n \log n/915)^{1/14.3}$, where the finance survivors ($L \approx n^{0.59}$) live.

These statements are not interchangeable. Theorem 4.4 is the conceptual sharp inequality (constant 1). Corollary 4.5 is the convenient length-only statewise bound that turns a descent floor into a per-length exclusion. Theorem 4.6 is the numerical certification of that bound, and it uses a conservative coefficient 6/5 only as a uniform majorant; the headline cutoff 25781 is not an artifact of that majorant. Theorem 4.7 is a run-type refinement of the same defect sum. The leftover lengths through 10^5 are supporting material, not a second main theorem.

1.1 Related work

Pickover's later exposition is Chapter 45 of [2], already cited above. Weisstein [5] records the map, the stopping-time sequences A094670, A094679, A095908, and a verification of arrival at 1 through 10^6. That verified descent floor is the computational input to Theorem 4.6; the first-passage run of Appendix B recomputes it. OEIS A094716 [6] records extreme heights, including the start 48443 whose peak has 972 463 digits. Those height records do not bound the period.

Prasad–Prasad [7] estimate excursion and stopping constants for juggler-like maps by a random-walk large-deviation model; those estimates do not apply to exact cycles. Small-cycle censuses are a standard first layer for Collatz-like maps, surveyed by Lagarias [8,9]. Those results do not transfer: the branches of J are floor powers rather than affine maps (Crandall [10], Matthews–Watts [11]). In particular there is no identity of the form $n(2^K - 3^p) = C$. A check of Pickover [1,2], Weisstein [5], the OEIS records [3,4,6], the Prasad–Prasad estimates [7], and the standard Collatz cycle-bound sources [8–13] found no published explicit lower bound on the period of a nontrivial cycle for this exact floor-power Juggler map. Informal near-miss notes and later unverified webpage floors are not used.

Goodstein's theorem is a superficially similar statement about natural numbers — every Goodstein sequence terminates at 0 — which cannot be proved in Peano arithmetic [14]. No such independence claim is made for the Juggler conjecture, and this paper does not study Goodstein sequences.

Companion manuscripts. Two later manuscripts by the author build on this one and are cited where they bear on the cycle problem. Paper B [16] proves parity equidistribution of the nested floor powers along the itineraries of J, with power savings, completely through depth four for odd-rooted itineraries and for the two length-five contractors, giving the certified-descent densities 13/16 and 7/8; its only input from this paper is the contraction criterion of Theorem 2.2. Paper C [17] proves that every nonempty backward-closed set — in particular the basin of any nontrivial cycle, and the set of divergent starts — has logarithmic count $\gg (\log x)^\lambda$ for $\lambda < 0.4050$, and reduces the Juggler conjecture to a Tao-type almost-all statement whose bounded target is the certified floor of Section 5 and whose descent step is the power envelope of Theorem 2.2 (`power_bound_word`). Section 6.1 states precisely what those results add to the cycle problem, and what they leave untouched.

The layers of the argument are as follows.

1. *Classical ideas, not claimed as new:* cycle financing (Simons–de Weger [12]); logarithmic and continued-fraction approximation of $\log 2/\log 3$, including Rhin's effective measure for $o \log 3 - L \log 2$ [15], used only in Corollary 4.11; cycle-itinerary restrictions and leftover packaging (Eliahou [13]; Lagarias [8,9]); the Denjoy–Koksma inequality and Ostrowski numeration, used as known tools in Section 5.
2. *New object:* the exact one-step floor-power preimages of J.
3. *New theorem:* the Juggler-specific cycle-minimum finance inequality of Theorem 4.4.
4. *New consequence:* $L \geq 25781$ at the verified descent floor 10^6.
5. *Supporting organization:* the run-packing refinement of the same defect sum (Theorem 4.7).
6. *New theorem:* the reduced-base transport and the uniform window envelope of Section 5 (Theorems 5.3 and 5.8).
7. *New consequence:* period at least 176251 at the laboratory floor 26254995 (Theorem 5.9), 478245 at the second certified floor 162849448 (Corollary 5.10), and 780239 at the third certified floor 350000000 (Corollary 5.11), every survivor inside the window of Theorem 5.8.
8. *Reduction:* the floor-free gap transfer (Theorem 4.10) and the short-cycle exclusion $L^{14.3} > n \log n/915$ (Corollary 4.11), which locate the open problem in the long regime without claiming anything there.

The argument below is elementary and independent of the Diophantine tools of [12]: floor-power defects are relatively $O(1/x)$ in logarithms, so a uniform logarithmic floor-error bound, valid above the verified floor, excludes every length that is not a finance-survivor for Theorem 4.4. To the best of our knowledge, the Juggler-specific inequality and the explicit period bound it produces are new.

Novelty statement. For this nonlinear floor-power map, exact floor defects convert into a cycle-financing inequality that forces the minimum of a hypothetical cycle below the independently verified descent region unless the period is at least 25781. Coupling the states through the closed exponent walk and bounding the extremal rotation itinerary by Denjoy–Koksma over certified Ostrowski blocks raises that period bound to 176251 at the laboratory floor, census-free on an explicit window of lengths, and — by certified evaluation of the same kill criterion on the survivors beyond that window — to 478245 at the second certified floor and

780239 at the third.

1.2 Verification

The arguments of Sections 2, 3, and 4 may be read without machine assistance. The core mathematical lemmas are mechanized in Lean 4; selected finite classifications and numerical tables are independently certified computations. Appendix A records the Lean names. The finite tables used by Lemmas 3.5, 3.7, 3.11 and Theorems 3.12–3.20 are `native_decide` evaluations in the modules named there.

Every number printed in Sections 4 and 5 is additionally recomputed from the *printed* criterion, independently of the probes that produced the tables, by `research.juggler_sequence.paper_a_audit`: the record $n_{\max}$ values, the contiguous excluded prefix at each of the four certified floors, the Rhin constants of Corollary 4.11, the convergent asymptotic of Section 4, and the fan law of Proposition 5.12. Two of those recomputations are delicate enough to be worth stating. The crossings that define $n_{\max}(25781)$ and $n_{\max}(50508)$ are sharp to a relative $2 \cdot 10^{-8}$ and $3 \cdot 10^{-10}$ respectively — each is the *last* integer at which the parity comparison holds, and the next one fails — so $heta = 1 - 2^L/3^o$ must be evaluated in extended precision: a double-precision evaluation of the exponent $L \log 2 - o \log 3$ carries enough relative error to move $n_{\max}(25781)$ to 26254996. The remaining terms may be evaluated in double precision, and are.

Proposition 1.3 (certified computational input). A machine-verifiable certificate establishes that every integer $2 \leq n \leq 10^6$ reaches 1. Precisely: an exact-integer first-passage run records, for each such n, a finite realized word with image strictly below the start; strong induction on that image reaches 1. The longest first passage in the window has 253 steps (seed 78901). Weisstein [5] records the same computational verification; the run here is an independent recomputation. The certificate files, SHA-256 hashes, and regeneration commands are Appendix B.

Roles. Independently proved: the finance inequality (Theorem 4.4). Computational input: every $2 \leq n \leq 10^6$ reaches 1 (this proposition), every $2 \leq n \leq 26254995$ reaches 1 (Proposition 5.1, the laboratory instance), every $2 \leq n \leq 162849448$ reaches 1 (Corollary 5.10, the second laboratory instance), and every $2 \leq n \leq 350000000$ reaches 1 (Corollary 5.11, the third laboratory instance). Independently recomputed: the exact first-passage runs of Appendix B. Not proved: global termination.

Theorem 4.6 applies Corollary 4.5 to this input and certifies the table with the conservative coefficient 6/5; its certified identity is Lean (`cycleMin_defect_finance`, `DefectFinance.lean`), the per-length table stays a verified computation. Theorem 4.7 is a human proof; Theorem 4.8 reuses the gap table under that packing. Proposition 4.9 is integer arithmetic in Lean. In Section 5, Theorem 5.7 is a human proof (Denjoy–Koksma is used as a known tool; its per-block hypotheses — convergent quality $|\theta - p/q| < 1/q^2$ and the block permutation — are Lean, `theta_convergent_quality`, `theta_block_permutations`) and so is the rotation identification in Lemma 5.6, whose itinerary identity itself is Lean (`budgetedWord_eq_hugWord`); the Laplace bound of Proposition 5.5 is Lean (`rotation_average_le`, `rotationAverage_gap`; the ergodic identification stays human); the transport inequality of Theorem 5.3 (`cycleMin_transport`), the defect-to-hug-charge consequence of §5.2 (`cycleMin_defect_le_charge`, `cycleMin_defect_le_hug_charge`), the charge-maximisation half of Theorem 5.4 (`hug_charge_maximal`; strict uniqueness stays human), the digit-cap step and digit scan of Theorem 5.8 (`ostroDigit_le`, `theta_digitSum_le`, `window_digit_scan`), and the kill template of Theorem 5.9 (`cycleMin_hug_kill_criterion`) are Lean-verified; Theorems 5.2 and 5.9 are independently certified computations at the laboratory floor, with the same trust boundary as Theorem 4.6 (exact integer arithmetic plus guarded float comparisons). This is not a claim that the paper as a whole is formally verified.

```
Repository:  https://github.com/sneakyweasel/balanced_ternary
Commit:      7802f78bec58c68cec92a2efb3db4a502f916277
Lean:        leanprover/lean4:v4.33.0
Mathlib:     v4.33.0 (lake-manifest rev db584cd6d46c92f209a44c0f1c829460d327499d)
Build:       lake build Problems.JugglerPaper   (from formal/)
Computation: python -m research.juggler_sequence.cycle_finance
             (parity table); run-type table in budget_opt.json
SHA-256:     Appendix B
```

The commit is the repository state that produced the finance tables. A later editorial commit of this text does not change those hashes.

2. Envelope

Let $\mathcal{B} = \{E, O\}$. A finite itinerary $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the trajectory of n are exactly the letters of w. Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters. When w is realized, this endpoint is the $|w|$-fold iterate.

Theorem 2.1 (fixed-itinerary monotonicity). If $n \le m$ and both realize w, then $J^{|w|}(n) \le J^{|w|}(m)$.

Proof. Induct on w. The empty itinerary is immediate. The images after a common realized prefix remain ordered, and both current states realize the same next letter. The even branch is the monotone integer square root; the odd branch is $x \mapsto x^3$ followed by the integer square root. □

The realizing set of a fixed word need not be an interval: OE is realized at 7 and at 11 but not at 9.

Theorem 2.2 (finite-itinerary power envelope). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \le n^{3^{\#O(w)}}.$$

Proof. The empty itinerary is the equality $n \le n$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $x^{2^\ell} \le n^{3^o}$, and the next letter is realized.

If the next letter is even, then $J(x)^2 \le x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \le x^{2^\ell} \le n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x)^2 \le x^3$, so

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \le x^{3 \cdot 2^\ell} = (x^{2^\ell})^3 \le \left(n^{3^o}\right)^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. □

Corollary 2.3 (exponent-gap contraction). If $n \ge 2$, w is realized at n, and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \le n^{3^{\#O(w)}}$. The exponent gap and $n \ge 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \ge 1$, one has $m < n$. □

The corollary includes familiar contracting blocks such as $OOOEE$. It does not prove that every start realizes some contracting itinerary.

2.4 Exact floor defect

Theorem 2.2 is the inequality form of an exact identity

$$n^{3^{\#O(w)}} = J^{|w|}(n)^{2^{|w|}} + \Delta_w(n), \qquad \Delta_w(n) \ge 0.$$

The identity, its vanishing law, and the two-term composition are Theorems 2.4–2.6 and Corollary 2.7 in Appendix C. A mixed realized word has $\Delta_w(n) > 0$. This explains why the envelope of Theorem 2.2 is true, and why a mixed cycle itinerary is formally expanding. Theorem 4.4 uses only the nonnegativity $\Delta_w(n) \ge 0$.

No uniform local tax exists: the relative slack of a single letter tends to 0 with the state. Finance must therefore be a global comparison, not a fixed cost per odd or even step. A quantitative itinerary-dependent lower bound on Δ_w, or a tighter upper bound on $\sum 1/(x_i \log x_i)$, would improve the period cutoff; neither is proved here.

3. Structural restrictions on cycle itineraries

The role of this section in the paper is structural, not numerical. The minimum geometry established here — the cycle minimum is odd, prefixes to even states are superquadratic (Theorem 3.2), a minimum-based itinerary has the canonical run form of Lemma 3.21b, and the last even letter lands in an explicit one-step preimage (Lemma 3.4(iv)) — is exactly what the finance unroll of Section 4 and the transport of Section 5 consume. The even-count exclusion (Theorem 3.22: every nontrivial cycle itinerary has at least four even letters, hence period at least eleven) is the section's own headline, but as a period bound it is superseded the moment financing appears; it is retained because it is floor-free. The small-period censuses (Theorems 3.6 and 3.8) are supporting. Main-text proofs are kept to the short structural lemmas; the longer case analyses — Lemmas 3.5 and 3.7, the censuses of Theorems 3.6 and 3.8, and the family exclusions of Theorems 3.12–3.21 — are Appendix D. After the one-step preimages, a minimum-based itinerary has a canonical run form. There are only three even-count regimes with $e \le 3$; each reduces to a finite collection of geometries $O^aEO^bEO^cE$. Those geometries are eliminated by a next-square obstruction, by long odd-run growth against a last-even one-step preimage, or by a finite exceptional window. The family calculations are Appendix D.

The exact one-step fibers are

$$J(n) = q \iff q^2 \le n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \le n^3 < (m+1)^2 \quad (n \text{ odd}).$$

Lemma 3.1 (odd one-step preimages are unique). An odd fiber contains at most one integer. An even fiber is a parity-restricted square interval and may contain many predecessors.

Proof. Suppose $a < b$ lie in the same odd one-step preimage indexed by m. Then $(a+1)^3 \le b^3 < (m+1)^2$ and $m^2 \le a^3$. Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a+1)^3 - a^3 < (m+1)^2 - m^2 = 2m + 1,$$

hence $3a^2 + 3a < 2m$ and $3a^2 < 2m$. If $a = 0$, then $m = 0$ and the displayed strict inequality is already impossible. If $a > 0$, squaring and using $m^2 \le a^3$ gives $9a^4 < 4m^2 \le 4a^3$, contradicting $4a^3 < 9a^4$. □

Theorem 3.2 (cycle restrictions). Let w be a cycle itinerary at $n \ge 2$.

(i) The itinerary is formally expanding:

$$2^{|w|} < 3^{\#O(w)}.$$

A contracting itinerary cannot close a nontrivial cycle.

(ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter.

(iii) A realized itinerary v is *superquadratic* if

$$3^{\#O(v)} \ge 2^{|v|+1}.$$

Any realized path from a start $n \ge 2$ to a state at least n^2 is superquadratic. On a cycle minimum the path to any later even state is superquadratic. The prefix OOE is expanding ($9 > 8$) but not superquadratic ($9 < 16$), so it cannot carry the minimum to square scale.

Proof. For (i), Theorem 2.2 applied to a cycle endpoint gives $n^{2^{|w|}} \le n^{3^{\#O(w)}}$. Since $n \ge 2$, comparison of exponents gives $2^{|w|} \le 3^{\#O(w)}$; equality is impossible because the two sides have different prime divisors and $|w| \ge 1$. Alternatively: a mixed cycle itinerary has $\Delta_w(n) > 0$ by Theorem 2.5 in Appendix C, so the envelope is strict; a monochrome tower cannot return for $n \ge 2$.

Every state on such a cycle is at least 2: once a trajectory reaches 1, it remains there and cannot return to a start $n \ge 2$. An even state $x \ge 2$ satisfies $J(x) < x$, so a cycle minimum cannot be even. An odd state $x \ge 3$ satisfies $J(x) > x$, so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let x be the predecessor of a minimum-oriented return n. If the last letter were odd, the odd return one-step preimage would give $n^2 \le x^3 < (n+1)^2$. Minimality gives $x \ge n$, hence $n^3 < (n+1)^2$, impossible for $n \ge 3$; and the minimum is odd, so $n \ge 3$.

For (iii), let a realized itinerary v send n to $y \ge n^2$. Theorem 2.2 gives

$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \le y^{2^{|v|}} \le n^{3^{\#O(v)}}.$$

Since $n \ge 2$, comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state y is even, its successor is at least n. Thus $n \le J(y) = \lfloor \sqrt{y} \rfloor$, so $n^2 \le y$, and the preceding argument applies to that prefix. ☐

Lemma 3.21b (canonical run form). After rotation to a minimum-based orientation, the itinerary begins with OO and ends with E; hence every itinerary with $1 \le e \le 3$ even letters has the canonical run decomposition $O^aEO^bEO^cE$ (unused runs empty). The case $e = 0$ is all-odd and is already forbidden by the last-letter restriction. No other cyclic rotation needs a separate case.

Proof. Theorem 3.2: the minimum is odd, so the itinerary cannot begin with E or OE, and it cannot end with an odd letter. Thus a minimum-based itinerary starts OO and ends E, so $e \ge 1$. The remaining letters are odd runs separated by the e even letters, so the itinerary is $O^{a_1}E \cdots O^{a_e}E$ with $a_1 \ge 2$. For $e \le 3$ this is $O^aEO^bEO^cE$ with unused runs empty. Every cycle itinerary has a minimum-based rotation of the same even-count, and that orientation is already in this form. ☐

Thus it is enough to exclude the three regimes $e = 1, 2, 3$. The rest of the section does that. Theorems 3.6 and 3.8 record the short-period consequences; Theorem 3.22 is the structural statement.

Lemma 3.3 (coarse lower envelope). For $n \ge 1$, write $q = \lfloor \sqrt{n} \rfloor$. Then $q^2 \le n < (q+1)^2$. For $q \ge 1$ one has $(q+1)^2 \le 4q^2$, and the second inequality is strict for $q \ge 2$, while for $q = 1$ one has $n < 4$. In all cases

$$n < 4 \lfloor \sqrt{n} \rfloor^2.$$

Thus an even step satisfies $n \le 4\, J(n)^2$ and an odd step satisfies $n^3 \le 4\, J(n)^2$. Composing along a realized itinerary v of length k with odd count o gives

$$n^{3^o} \le C_v\, J^k(n)^{2^k}.$$

The constant starts at 1 and updates by $C \mapsto C \cdot 4^{2^j}$ on an even letter and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j. In particular $C_{OOO} = 2^{38}$ and $C_{OOOO} = 2^{130}$.

Proof. The comparison $n < 4q^2$ is the paragraph above. The composition law is the same recurrence as the proof of Theorem 2.2, with a factor 4^{2^j} inserted at each letter. The values C_{OOO} and C_{OOOO} are the result of that recurrence on those two words. ☐

Lemma 3.4 (next-square thresholds). (i) If $q \ge 5$ realizes OO, then $J^2(q) \ge (q+1)^2$. (ii) If $q \ge 3$ realizes OOO, then $J^3(q) \ge (q+1)^2$. (iii) If a realized itinerary v satisfies $J^{|v|}(q) \ge (q+1)^2$ and the next realized letter is odd, then $J^{|v|+1}(q) \ge (q+1)^2$. (iv) If vE is a cycle itinerary at n, then $J^{|v|}(n) < (n+1)^2$. (v) If $a \ge 3$, then O^aE is not a cycle itinerary at any $n \ge 2$.

Proof. For (i), write $m = \lfloor q^{3/2} \rfloor$. The bound $J^2(q) \ge (q+1)^2$ follows from $m^3 \ge (q+1)^4$, because then $\lfloor m^{3/2} \rfloor \ge (q+1)^2$. If $q = 5$, then $11^2 = 121 \le 125 = 5^3 < 144 = 12^2$, so $m = 11$, and $11^3 = 1331 \ge 6^4 = 1296$. Since q realizes OO, it is odd, so the only remaining case is $q \ge 7$. Then $m \ge q^{3/2} - 1$, so it is enough that $(q^{3/2} - 1)^3 \ge (q+1)^4$. Expanding the left side and dropping the positive remainder $3q^{3/2} - 1$ reduces this to $q^{9/2} - 3q^3 \ge (q+1)^4$, or equivalently $\sqrt{q} - 3/q \ge (1 + 1/q)^4$. The left side increases for $q > 0$ and the right side decreases, so the case $q = 7$ suffices:

$$\sqrt{7} - \frac{3}{7} > \frac{5}{2} - \frac{3}{7} = \frac{29}{14} > \frac{4096}{2401} = \left(\frac{8}{7}\right)^4,$$

where $\sqrt{7} > 5/2$ follows from $25/4 < 7$.

For (ii), the trajectory $3 \to 5 \to 11 \to 36$ gives $J^3(3) = 36 \geq 16$. If $q \geq 5$ realizes OOO, then it realizes OO, so (i) gives $J^2(q) \geq (q+1)^2$. The third letter is odd, hence $J^3(q) = J(J^2(q)) \geq J^2(q)$.

For (iii), the image after v is odd and at least $(q+1)^2 \geq 4$, so the odd branch does not decrease it.

For (iv), the last letter is even, so the preimage $z = J^{|v|}(n)$ is even and satisfies $n^2 \leq z < (n+1)^2$.

For (v), suppose $O^a E$ is a cycle itinerary at $n \geq 2$. The start is odd, hence at least 3, and realizes O^a. Parts (ii) and (iii) give $J^a(n) \geq (n+1)^2$, contradicting (iv). $\square$

Lemma 3.5 (two length-six exclusions). Neither *OOOEOE* nor *OOOOEE* is a cycle itinerary at any $n \geq 2$.

Proof. Appendix D.

Theorem 3.6 (small-cycle census). No itinerary of length at most six is a cycle itinerary at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

Proof. Appendix D. The reduction used there — an all-odd itinerary cannot return, and every mixed cycle itinerary rotates to an even-terminating cycle itinerary based at a cycle state $m \geq 2$ — is reused by Theorem 3.8 and Theorem 3.22.

Lemma 3.4(v) excludes every odd-run-then-even word $O^a E$ with $a \geq 3$, of any length.

Lemma 3.7 (two length-seven exclusions). Neither *OOOOEOE* nor *OOOOOEE* is a cycle itinerary at any $n \geq 2$.

Proof. Appendix D.

Theorem 3.8 (small-cycle census through length seven). No itinerary of length at most seven is a cycle itinerary at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least eight.

Proof. Appendix D.

Theorems 3.6 and 3.8 are supporting period statements. The structural claim is the even-count exclusion of Theorem 3.22. The same one-step preimages organise leftover itineraries by even-count. Write

$$e_a = 2(3^a - 2^a)$$

for $a \geq 0$.

Lemma 3.9 (trailing even run). If a cycle itinerary based at n ends with $r \geq 1$ even letters, the state immediately before that run is strictly less than $(n+1)^{2^r}$.

Proof. The case $r = 1$ is Lemma 3.4(iv). Suppose the claim holds for some $r \geq 1$, and let vE^{r+1} be a cycle itinerary at n. Write $z = J^{|v|}(n)$. The inductive hypothesis applied to the suffix E^r after the first of those even letters gives $J(z) < (n+1)^{2^r}$. The state z is even, so $z < (J(z)+1)^2 \leq \left((n+1)^{2^r}\right)^2 = (n+1)^{2^{r+1}}$. $\square$

Lemma 3.10 (odd-run lower envelope). If $n \geq 1$ realizes O^a, then

$$n^{3^a} \leq 2^{e_a} \, J^a(n)^{2^a}.$$

Proof. Lemma 3.3 supplies a multiplicative constant C_v along any realized itinerary, with $C_\varepsilon = 1$, $C \mapsto C \cdot 4^{2^j}$ on an even letter, and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j. On the pure odd word O^a every letter is odd, so $C_{O^{a+1}} = C_{O^a}^3 \cdot 4^{2^a}$. Writing $C_{O^a} = 2^{e_a}$ with $e_0 = 0$ yields the recurrence $e_{a+1} = 3e_a + 2^{a+1}$, because $4^{2^a} = 2^{2^{a+1}}$. The closed form $e_a = 2(3^a - 2^a)$ satisfies the recurrence and the initial value. $\square$

Lemma 3.11 (seven-odd window). No integer n with $2 \leq n < 256$ realizes the itinerary O^7.

Proof. This is a table of 254 seven-step evaluations: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_follows_seven_odds_of_lt256` (Appendix A). $\square$

Theorem 3.12 (two-even leftover families). Let $k \geq 6$ and $n \geq 2$. Neither $O^{k-2}EE$ nor $O^{k-3}EOE$ is a cycle itinerary at n.

Proof. Appendix D.

A cycle itinerary w at n is *minimum-based* when n is a cycle minimum: $J^j(n) \geq n$ for every $0 \leq j < |w|$. The remainder after a proper prefix of a cycle itinerary need not itself be a cycle itinerary at the prefix endpoint. The next statement therefore transports the tail inequality of Theorem 3.12, not the cycle-itinerary exclusion at a later start.

Theorem 3.13 (first-even transport). Let $n \geq 2$. No minimum-based cycle itinerary at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Appendix D.

The hypothesis that the start is a cycle minimum is essential. If $y < n$, the leftover one-step preimage is measured against a larger start and need not contradict the tail at y. In particular, Theorem 3.13 does not assert that those itineraries fail to be cycle itineraries at a non-minimum start. That upgrade is Theorem 3.21.

Theorem 3.14 (three trailing evens). Let $a \geq 6$ and $n \geq 2$. The itinerary O^aEEE is not a cycle word at n.

Proof. Appendix D.

Theorem 3.15 (mixed bunched family $EOEE$). Let $a \geq 5$ and $n \geq 2$. The itinerary O^aEOEE is not a cycle word at n.

Proof. Appendix D.

Theorem 3.16 (mixed bunched family $EOOEE$). Let $a \geq 4$ and $n \geq 2$. The itinerary O^aEOOEE is not a cycle itinerary at n.

Proof. Appendix D.

Theorem 3.17 (mixed bunched family $EOOOEE$). Let $a \geq 3$ and $n \geq 2$. The itinerary $O^aEOOOEE$ is not a cycle itinerary at n.

Proof. Appendix D.

Theorem 3.18 (mixed bunched family $EEOE$). Let $a \geq 5$ and $n \geq 2$. The itinerary O^aEEOE is not a cycle itinerary at n.

Proof. Appendix D.

Theorem 3.19 (mixed bunched family $EOEOE$). Let $a \geq 4$ and $n \geq 2$. The itinerary O^aEOEOE is not a cycle itinerary at n.

Proof. Appendix D.

Theorem 3.20 (mixed bunched family $EOOEOE$). Let $a \geq 3$ and $n \geq 2$. The itinerary $O^aEOOEOE$ is not a cycle itinerary at n.

Proof. Appendix D.

Theorem 3.21 (gapped leftovers as cycle itineraries). Let $n \geq 2$. No cycle itinerary at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Appendix D.

Theorems 3.12–3.21 assemble into an even-count exclusion: no cycle itinerary has fewer than four even letters, so a nontrivial cycle has period at least eleven (Theorem 3.22). Section 4 excludes later periods by financing.

Every integer $2 \leq n < 12$ reaches 1, so a cycle minimum is at least 12. Write e for the number of even letters. Lemma 3.21b is the canonical run form.

Lemma 3.21a (classification). Every minimum-based cycle itinerary with at most three even letters belongs to one of the families excluded by Lemma 3.4(v) and Theorems 3.12–3.21.

Proof. Lemma 3.21b puts the itinerary in the form $O^aEO^bEO^cE$ with $a \geq 2$ and unused runs empty. If $e = 0$, the itinerary is all-odd. If $e = 1$, it is O^aE. If $e = 2$, a last run $c \geq 2$ is the internal-even bootstrap of Lemma 3.4, and the remaining shapes are the two-even families of Theorem 3.12. If $e = 3$, again $c \geq 2$ is the bootstrap, while $c \in \{0, 1\}$ is either a gapped leftover (Theorem 3.21) or one of the seven bunched families (Theorems 3.14–3.20). □

e	remaining minimum-based forms	elimination
0	all-odd	cannot return (Theorem 3.6)
1	O^aE	next-square; Lemma 3.4(v)
2	O^aEE, O^aEOE, last run ≥ 2	Theorem 3.12; Lemma 3.4
3	bunched families and gapped leftovers	Theorems 3.14–3.21
≥ 4	not excluded by even-count	finance (Section 4)

Theorem 3.22 (even-count). No itinerary with fewer than four even letters is a cycle itinerary at any $n \geq 2$. Equivalently, a nontrivial cycle itinerary has at least four even letters.

Proof. Every cycle itinerary has a minimum-based rotation, with the same even-count. Lemma 3.21b puts that orientation in canonical run form; Lemma 3.21a names the families. In detail:

If $e = 0$, the itinerary is all-odd and cannot return, as in Theorem 3.6.

If $e = 1$, the itinerary is O^aE with $a \geq 2$. The case $a = 2$ is OOE, excluded in Theorem 3.6. The cases $a \geq 3$ are Lemma 3.4(v).

If $e = 2$, the itinerary is O^aEO^cE with $a \geq 2$. A last odd-run $c \geq 2$ is an internal even letter followed by OO or OOO. Lemma 3.4 at the cycle minimum ($n \geq 12$) contradicts the last-even one-step preimage. The remaining shapes are O^aEE and O^aEOE. Expansion forces $a \geq 4$ and $a \geq 3$ respectively, so both are Theorem 3.12.

If $e = 3$, the itinerary is $O^aEO^bEO^cE$ with $a \geq 2$. Again $c \geq 2$ is the internal-even bootstrap. The remaining last runs are $c = 0$ and $c = 1$. For $c = 0$ and $b \geq 4$ the itinerary is a gapped leftover O^aEO^bEE, excluded by Theorem 3.21. For $c = 0$ and $b \leq 3$ the word is one of O^aEEE, O^aEOEE, O^aEOOEE, $O^aEOOOEE$, excluded by Theorems 3.14–3.17 once expansion supplies the stated lower bounds on a. For $c = 1$ and $b \geq 3$ the itinerary is a gapped leftover O^aEO^bEOE, excluded by Theorem 3.21. For $c = 1$ and $b \leq 2$ the itinerary is one of O^aEEOE, O^aEOEOE, $O^aEOOEOE$, excluded by Theorems 3.18–3.20.

Thus $e \leq 3$ is impossible. □

Corollary 3.23. A nontrivial cycle, if one exists, has period at least eleven.

Proof. Theorem 3.22 gives four even letters, hence $o \leq L - 4$. Formal expansion is $2^L < 3^o \leq 3^{L-4}$. The comparison $2^L < 3^{L-4}$ first holds at $L = 11$. □

In particular there is no cycle of length eight, nine, or ten.

4. Cycle finance

A cycle itinerary is formally expanding (Theorem 3.2), yet the trajectory returns exactly. The multiplicative surplus $3^o - 2^L$ must be financed by the floor remainders, which are relatively $O(1/x)$ in logarithms. The resulting bound on the cycle minimum excludes every period that is not admissible for the inequality, once a uniform logarithmic floor-error bound is available above a verified descent floor.

The identity is unrolled along a circular word. After rotation to a cycle minimum that itinerary is a necklace of odd-run excursions. The geometry below records that itinerary. It introduces no new theorem: every named constraint is Theorem 3.2, Lemma 3.4, Lemma 3.21b, or the last-even one-step preimage.

The excursion necklace

Write n for a cycle minimum and

$$w = O^{a_1} E O^{a_2} E \cdots O^{a_e} E$$

for a minimum-based orientation (Lemma 3.21b), so $a_1 \geq 2$, $\sum_i a_i = o$, and $e = L - o$. The odd landings are the *valleys* v_i and the even state just before each final E is the *peak* p_i:

$$v_0 = n, \qquad p_i = J^{a_{i+1}}(v_i), \qquad v_{i+1} = J(p_i) = \lfloor \sqrt{p_i} \rfloor \quad (0 \leq i < e),$$

with $v_e = n$. The itinerary is

$$n \xrightarrow{OO} \text{first high region} \xrightarrow{E} v_1 \xrightarrow{O^{a_2} E} \cdots \xrightarrow{O^{a_{e-1}} E} v_{e-1} \xrightarrow{O^{a_e} E} n.$$

Two meanings of *entry* must not be conflated. *Cycle entry* is the distinguished cut that places the minimum at the start of the itinerary. *Dynamical entry* is the last even step into n. The second is a genuine boundary condition; the first is a choice of origin.

Cycle minimum and the forced lift The minimum is odd, so the itinerary cannot begin with E or OE, and it cannot end with an odd letter (Theorem 3.2). The first two letters are therefore OO:

$$n \xrightarrow{O} y \xrightarrow{O} z.$$

For $n \geq 5$ one has $z = J^2(n) \geq (n+1)^2$ (Lemma 3.4(i)). In particular OOE cannot be a cycle itinerary (`no_cycle_itinerary_ooe`). On a cycle minimum the first even residual overshoots the entry one-step preimage: the first peak satisfies $p_0 \geq (n+1)^2$. That is the opposite of the last-peak condition below. Even $J^2(n)$ may continue with E; odd $J^2(n)$ continues with O. Either way the minimum-based prefix is still OO.

Excursions, valleys, and peaks An ordinary excursion is one block $O^a E$. In the itinerary semantics that itinerary is `oddEvenBlock a 1`. Write

$$\mu(a) = \frac{3^a}{2^{a+1}}$$

for the ideal (floor-free) exponent of the block. The block is formally expanding if and only if $\mu(a) > 1$, equivalently $2^{a+1} < 3^a$. Thus OE contracts ($\mu(1) = 3/4$) and OOE expands ($\mu(2) = 9/8$). This is a reparameterization of the itinerary envelope of Theorem 2.2, not a transition law on pairs (a_i, a_{i+1}). Lemma 3.4(v) forbids $O^a E$ as a *cycle itinerary* for $a \geq 3$; it does not forbid an internal block of that shape.

The trajectory is then the wave

$$v_0 \to p_0 \to v_1 \to p_1 \to \cdots \to v_{e-1} \to p_{e-1} \to v_0.$$

On a cycle minimum every even state is already at least n^2 (Theorem 3.2(iii)). Valleys dominate a logarithmic defect sum because $1/(x \log x)$ is largest there; peaks are huge and cheap. That is why the run-type packing of Theorem 4.7 charges `OOE`-scale valleys, `OE`-scale valleys, and evens separately. The packed comparison is an extremal charge of the same sum, not a uniqueness theorem for the actual word. Expanding blocks can climb and a later OE can drop without crossing the anchor: four consecutive expanding blocks occur already at the certified start 1999 recorded in Section 6.

Closure and the entry one-step preimage The valley sequence is circular. The last letter is E, so the last peak occupies the last-even one-step preimage of Lemma 3.4(iv):

$$n^2 \le p_{e-1} < (n+1)^2.$$

The minimum is odd, so $p_{e-1} \ne n^2$ (`cycle_last_even_ne_odd_sq`) and p_{e-1} is even. Thus

$$n^2 + 1 \le p_{e-1} < (n+1)^2, \qquad p_{e-1} \text{ even}.$$

An ordinary excursion needs only $v_i \ge n$. The last excursion must hit this one-step preimage and land on n:

$$v_{e-1} \xrightarrow{O^{a_e}} p_{e-1} \xrightarrow{E} n.$$

The first peak overshoots the same one-step preimage; the last peak lands in it. Those are different even states.

Once the trajectory returns to n, the prefix OO is forced again. A genuine cycle is a closed necklace of excursions whose first block starts OO, whose last peak lies in the entry one-step preimage, and whose entire trajectory stays at least n. The global constraints already proved are

$$\sum_i a_i = o, \qquad e = L - o, \qquad 2^L < 3^o,$$

together with periodicity $\prod_{i=0}^{e-1} v_{i+1}/v_i = 1$.

What remains Finance (Theorem 4.4) sits around the necklace: the surplus $3^o - 2^L$ must be paid by floor losses, and at the verified floor $N_0 = 10^6$ this forces $L \ge 25781$. The run-type packing of Theorem 4.7 is a refinement of the same defect sum along the valleys and peaks just named. Subsequent refinements of one component of the necklace recovered Theorem 4.7 or closed. They are not claims of this note.

The pieces are understood separately: the forced lift at n, the blocks $O^{a_i}E$, and the last-even landing. What is not proved is a link strong enough to exclude the leftover lengths,

$$\text{minimum geometry} + \text{necklace of excursions} + \text{entry one-step preimage} \ \Rightarrow\ \perp.$$

That is a formulation of the remaining cycle problem, not a theorem. A genuine lower bound on the number of odd runs would feed Theorem 4.7; none is proved here.

Throughout this section write $L = |w|$ and $o = \#O(w)$ for a cycle itinerary w based at a cycle minimum $n \ge 2$. Natural logarithms are written log in this section and ln in Section 5; both denote the natural logarithm. The unique one-step fibres of Section 3 give, for every state $x \ge 1$ with image $y = J(x)$,

$$y^2 \le x^e < (y+1)^2, \qquad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}. \end{cases}$$

Lemma 4.1 (dyadic one-step-preimage logarithm). If $z, y \ge 1$ and $z < (y+1)^2$, then $\log z \le 2\log y + 2/y$.

Proof. The hypothesis gives $\log z \le 2\log(y+1)$. The inequality $\log(1+u) \le u$ for $u > 0$ yields $\log(y+1) = \log y + \log(1+1/y) \le \log y + 1/y$. □

Lemma 4.2 (one-step bounds). Let $x \ge 2$ and $y = J(x)$. If x is even, then $\log x \le 2\log y + 2/y$. If x is odd, then $3\log x \le 2\log y + 2/y$.

Proof. The even one-step preimage is $y^2 \le x < (y+1)^2$. The one-step-preimage logarithm lemma on $z = x$ is the first claim. The odd one-step preimage is $y^2 \le x^3 < (y+1)^2$. The same lemma on $z = x^3$ gives $\log(x^3) \le 2\log y + 2/y$. □

On a cycle minimum every proper prefix is non-contracting: a prefix with $3^{\#O} < 2^k$ would satisfy $J^k(n) < n$ by Corollary 2.3, contradicting minimality. Thus $2^k \le 3^{o_k}$ for every prefix of length k, where o_k is the odd count of that prefix.

Lemma 4.3 (unrolled envelope). Write $x_k = J^k(n)$ and o_k for the odd count of the length-k prefix. For every $0 \le k \le L$,

$$3^{o_k} \log n \le 2^k \log x_k + \frac{k\, 3^{o_k}}{n}.$$

Proof. The case $k = 0$ is an equality. Suppose the claim holds at $k < L$, and write $x = x_k$ and $y = x_{k+1}$. Minimality gives $n \le y$, and the prefix law gives $2^{k+1} \le 3^{o_{k+1}}$, hence

$$\frac{2^{k+1}}{y} \le \frac{3^{o_{k+1}}}{n}.$$

If the next letter is even, then $o_{k+1} = o_k$ and the one-step bound gives $\log x \le 2 \log y + 2/y$. Multiply by 2^k and add the inductive remainder:

$$3^{o_k} \log n \le 2^{k+1} \log y + \frac{2^{k+1}}{y} + \frac{k\, 3^{o_k}}{n} \le 2^{k+1} \log y + \frac{(k+1)\, 3^{o_k}}{n}.$$

If the next letter is odd, then $o_{k+1} = o_k + 1$. Multiply the inductive bound by 3 and apply the one-step bound in the form $3 \log x \le 2 \log y + 2/y$:

$$3^{o_k+1} \log n \le 2^{k+1} \log y + \frac{2^{k+1}}{y} + \frac{k\, 3^{o_k+1}}{n} \le 2^{k+1} \log y + \frac{(k+1)\, 3^{o_k+1}}{n}.$$

This is the claim at $k+1$. $\square$

Theorem 4.4 (finance). Let w be a cycle itinerary of length L with o odd letters, based at a cycle minimum $n \ge 2$. Then

$$n \log n \cdot (3^o - 2^L) \le L \cdot 3^o.$$

Proof. Apply Lemma 4.3 at $k = L$. Periodicity gives $x_L = n$ and $o_L = o$, so

$$3^o \log n \le 2^L \log n + \frac{L \cdot 3^o}{n}.$$

The itinerary is formally expanding, so $3^o > 2^L$. Rearranging and multiplying by n is the claim. $\square$

This is the conceptual centre of the note. Exact one-step preimages give a one-step logarithmic defect; cycle minimality lets the defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. Ideal dynamics expands and exact dynamics returns, so the floor errors finance the expansion. The inequality $\log(1+u) \le u$ is the only analytic input; the content is that interaction. The Lean form is exactly Theorem 4.4 (constant 1): `cycleMin_finance`.

Hierarchy of forms. These three layers must not be conflated.

1. *Theorem 4.4* is the conceptual sharp inequality. Constant 1:

$$n \log n \cdot (3^o - 2^L) \le L \cdot 3^o,$$

   equivalently $\theta \le L/(n \log n)$ with $\theta = 1 - 2^L/3^o$. It charges every one-step-preimage defect at the cycle minimum. Lean: `cycleMin_finance`.

2. *Corollary 4.4c* (below) is the same one-step-preimage-log unroll with remainders kept as $1/x_{i+1}$. It is the strongest proved form of those defects. Lean: `cycleMin_finance_inv_sum`.

3. *Corollary 4.5* is the convenient length-only statewise bound: a verified descent floor N_0 plus a parity charge of the defect sum produces a per-length threshold $n_{\max}(L)$ and excludes every L with $n_{\max}(L) \leq N_0$.
4. *Theorem 4.6* is the numerical certification of Corollary 4.5 at $N_0 = 10^6$. The certified identity is the conservative relative-defect form
$$1 - \frac{2^L}{3^o} \leq \frac{6}{5} \sum_{i=1}^{L} \frac{1}{x_i \log x_i}.$$
The factor 6/5 is a convenient uniform majorant of $-\log(1-\delta)/\delta$ on $[0, 1/6]$. It is not Theorem 4.4, and the headline cutoff 25781 is not an artifact of that majorant. The identity itself is Lean for any cycle minimum $n \geq 400$ (`cycleMin_defect_finance`, `DefectFinance.lean`); the per-length numeric table stays a verified computation.

The relative-defect unroll is a parallel identity, not Theorem 4.4 multiplied by 6/5. Charging every state at the cycle minimum in that identity recovers the coarser comparison $\theta \leq (6/5)L/(n \log n)$, which is Theorem 4.4 with coefficient 6/5. At this floor that uniform charge excludes only through length 1053. The computational table uses a stricter length-only parity charge of the same identity.

Corollary 4.4c (inv-sum). Let w be a cycle itinerary of length L with o odd letters, based at a cycle minimum $n \geq 2$. Write $x_i = J^i(n)$. Then
$$(3^o - 2^L) \log n \leq 3^o \sum_{i=1}^{L} \frac{1}{x_i},$$
equivalently $\theta \leq (\sum_i 1/x_i)/\log n$.

Proof. The same induction as Lemma 4.3, keeping each one-step preimage defect as $2^{k+1}/x_{k+1}$ instead of replacing it by $3^{o_{k+1}}/n$. At $k = L$ one has $x_L = n$. Lean: `cycleMin_log_envelope_inv`, `cycleMin_finance_inv_sum`. □

The computational table of Theorem 4.6 uses a weaker per-step bound, valid on every cycle because every start below 12 reaches 1 (so every cycle state is at least 12). For a state x with image $y = J(x) \geq 12$, the relative defect $\delta = (x^e - y^2)/x^e$ satisfies $\delta \leq 2/y \leq 1/6$. Writing $\varepsilon = -\frac{1}{2}\log(1-\delta)$ and using $-\log(1-\delta) \leq \frac{6}{5}\delta$ on $[0, 1/6]$ gives $\varepsilon \leq (6/5)/y$. Unrolling $t_{i+1} = (e_i/2)\, t_i - \varepsilon_i$ around the cycle then yields the conservative identity displayed in item 4 of the hierarchy. This chain is Lean end to end (`DefectFinance.lean`): the per-step losses in image form (`log_floorPower_even_ge_sub`, `log_floorPower_odd_ge_sub`, from $-\log(1 - \delta) \leq \frac{6}{5}\delta$ on $[0, 1/6]$, `neg_log_one_sub_le_sixth`), the amplification priced by the upper invariant $\log x_k \leq w_k \log n$ (`cycleMin_log_le_weight`), and the charged unroll (`cycleMin_charge_prefix`), closed at $x_L = n$.

On a minimum-based cycle of length L with o odd letters and $e = L - o$ even letters, the last letter is even, so $e \geq 1$ and the number of odd-run starts is at most e. Every even state satisfies $x \geq n^2$. Every odd state preceded by an odd state satisfies $x \geq t = \lfloor n^{3/2} \rfloor$. Therefore
$$\sum_{i=1}^{L} \frac{1}{x_i \log x_i} \leq \frac{e}{n \log n} + \frac{o - e}{t \log t} + \frac{e}{2n^2 \log n},$$
and
$$n \log n \left(1 - \frac{2^L}{3^o}\right) \leq \frac{6}{5} \left(e + (o - e)\frac{n \log n}{t \log t} + \frac{e}{2n}\right).$$
The optimal uniform coefficient on $[0, 1/6]$ is $6 \log(6/5) \approx 1.093$. Replacing 6/5 by that constant, or even by 1, on the same parity charge does not change the first surviving length: one still has $n_{\max}(25780) \leq 10^6 < n_{\max}(25781)$. The published table and the count 141 keep the proved coefficient 6/5. The cutoff 25781 is therefore not an artifact of an avoidable loss in the majorant.

Lemma 4.4b (odd-count monotonicity). Write $\theta(o) = 1 - 2^L/3^o$ and
$$R(o) = e + (o - e)\alpha + \frac{e}{2n}, \qquad e = L - o, \qquad \alpha = \frac{n \log n}{t \log t}.$$

The certified parity comparison is $n \log n \cdot \theta(o) \le (6/5)R(o)$. For every $n \ge 12$ one has $\alpha < 1/2$, hence

$$R(o+1) - R(o) = 2\alpha - 1 - \frac{1}{2n} < 0.$$

Also $\theta(o+1) > \theta(o)$. Therefore if the comparison fails at some admissible o, it fails at every larger odd count. Equivalently, the largest n at which the comparison can hold occurs at the least admissible odd count $o_{\min}(L) = \min\{o : 3^o > 2^L\}$. The same monotonicity holds for any positive coefficient in place of 6/5, and for the constant-1 comparison of Theorem 4.4.

Proof. The difference $R(o+1) - R(o)$ is the displayed coefficient of o. For $n \ge 12$ one has $t = \lfloor n^{3/2} \rfloor \ge n^{3/2} - 1 > 2n$, hence $\alpha < (n \log n)/(2n \log(2n)) = (\log n)/(2 \log(2n)) < 1/2$. The map $o \mapsto \theta(o)$ is strictly increasing on integers o with $3^o > 2^L$. If $n \log n \cdot \theta(o) > (6/5)R(o)$, then $n \log n \cdot \theta(o+1) > (6/5)R(o) > (6/5)R(o+1)$. □

Define

$$\gamma(L) = \frac{3^{o_{\min}(L)}}{2^L} - 1,$$

and write $n_{\max}(L)$ for the largest integer n at which the displayed parity inequality can still hold at $o = o_{\min}(L)$. The coarser comparison $n \log n \le (6/5)L \cdot 3^{o_{\min}}/(3^{o_{\min}} - 2^L)$ is used only as a check; Theorem 4.6 uses $n_{\max}(L)$ from the parity form. Lemma 4.4b is why that table may be computed at $o_{\min}$ alone.

Proposition 4.4a (finance-survivor algorithm). Fix a verified descent floor N_0. For each integer $1 \le L \le 10^5$, compute $o_{\min}(L) = \min\{o : 3^o > 2^L\}$ by exact integer arithmetic and $n_{\max}(L)$ from the parity inequality, and retain L if and only if $n_{\max}(L) > N_0$. The resulting finance-survivor set is

$$\mathcal{E}(N_0) = \{L : 1 \le L \le 10^5,\ n_{\max}(L) > N_0\}.$$

The printed instance is $\mathcal{E} = \mathcal{E}(10^6)$, with $|\mathcal{E}| = 141$.

Corollary 4.5. If every integer $2 \le n \le N_0$ reaches 1, then no nontrivial cycle of length L exists whenever $n_{\max}(L) \le N_0$.

Proof. A periodic state never reaches 1, so every cycle state is at least $N_0 + 1$. The length-only parity charge of the certified relative-defect identity then forces the minimum to satisfy the displayed inequality, hence $n \le n_{\max}(L)$. This is the convenient statewise bound in the hierarchy after Theorem 4.4; it is not a replacement for that theorem. □

Record values of the parity $n_{\max}$ include $n_{\max}(19) = 133$, $n_{\max}(84) = 2323$, $n_{\max}(569) = 23568$, $n_{\max}(1054) = 788014$, and $n_{\max}(25781) = 26254995$.

Theorem 4.6 (verified computation). Every integer $2 \le n \le 10^6$ reaches 1. Consequently:

(A) there is no nontrivial Juggler cycle of length at most 25780;

(B) if a nontrivial cycle has period $L \le 10^5$, then $L \in \mathcal{E}$, where $|\mathcal{E}| = 141$.

The bound closes continuously through 25780; 25781 is the first integer for which this particular inequality does not exclude a cycle. The coefficient 6/5 is the certification majorant of item 4 in the hierarchy after Theorem 4.4, not the source of the cutoff.

Proof. The descent floor is Proposition 1.3. The gap table computes $o_{\min}(L)$ and $n_{\max}(L)$ for every $1 \le L \le 10^5$ by Lemma 4.4b. Corollary 4.5 at $N_0 = 10^6$ excludes every L with $n_{\max}(L) \le 10^6$. The contiguous excluded prefix is $L \le 25780$. The complementary set in range is $\mathcal{E}$; checksums are Appendix B. □

Run packing

Theorem 4.7 refines the length-only charge by separating the unique minimum, `OOE`-scale valleys, `OE`-scale valleys, internal odds, and evens. It is a supporting comparison at the same floor; the leftover count $141 \to 99$ does not raise the cutoff. A cycle cannot put an `OE`-start at the minimum (Theorem 3.2), and an n-circuit of k odds and ℓ evens stays at least n only if $3^k \ge 2^{k+\ell}$.

Theorem 4.7 (run-type packing). Let w be a cycle itinerary of length L with $o = o_{\min}(L)$ odd letters and $e = L - o$ even letters, based at a cycle minimum $n \geq 12$. Write v for the least odd integer with $v^3 \geq n^4$, and write $t = \lfloor n^{3/2} \rfloor$. Then $o - e < e$, the largest number of n-scale valleys compatible with the even cap is $o - e$ copies of `OOE`, and the remaining $2e - o$ circuits are `OE` from v. The cycle minimum occurs once, so

$$\sum_{i=1}^{L} \frac{1}{x_i \log x_i} \leq \frac{1}{n \log n} + \frac{o-e-1}{(n+2)\log(n+2)} + \frac{2e-o}{v \log v} + \frac{1}{t \log t} + \frac{o-e-1}{t_+ \log t_+} + \frac{e}{2n^2 \log n},$$

where $t_+ = J(n+2)$. Combined with the 6/5 unroll this is strictly smaller than the parity sum of Corollary 4.5 whenever $2e - o > 0$. Sending the cycle maximum to infinity removes one even term and does not change the valley packing.

Proof. The statement is at $o = o_{\min}(L)$. Lemma 4.4b already excludes every larger odd count from the parity comparison used in Theorem 4.6; the packing is a refinement at that same odd count. Formal expansion at $o_{\min}$ forces $3o < 2L$ on every leftover length in range, equivalently $o - e < e$. The even-cap comparison $3^k \geq 2^{k+\ell}$ is the ideal power envelope; floors only help. An `OE`-start is followed by an even state, so Theorem 3.2 gives $J(v) \geq n^2$ and therefore $v^3 \geq n^4$. Unique visit of the cycle minimum is periodicity. The displayed sum charges one cheap valley at n, the remaining $o - e - 1$ cheap valleys at the next odd integer $n + 2$, the expensive valleys at v, one internal odd at t, the remaining internals at $J(n+2)$, and every even at n^2. Any deeper odd run or any higher valley only decreases the sum. □

Theorem 4.8 (run-type table). At the verified descent floor $N_0 = 10^6$, the packed comparison of Theorem 4.7 excludes the 42 lengths $56347 + 1054k$ for $k = 0, \ldots, 41$ and leaves an explicit set $\mathcal{E}_{\text{run}} = \mathcal{E}_{\text{run}}(10^6)$ of 99 lengths. The first survivor remains 25781. In particular, if a nontrivial cycle has period $L \leq 10^5$, then $L \in \mathcal{E}_{\text{run}}$.

Proof. The 141 lengths of Theorem 4.6(B) are tested at $n = 10^6 + 1$ against the packed right-hand side. The comparison is certified on each length: no entry is left uncertain. The 42 excluded lengths are exactly the arithmetic progression named in the statement. The complementary set in that range is $\mathcal{E}_{\text{run}}$. Checksums are Appendix B. The period cutoff remains 25781. □

Arithmetic structure of the finance survivors

The leftover lengths cluster around the continued-fraction approximants of $\log 2/\log 3$. That organizes the table; it does not constrain a hypothetical cycle.

Proposition 4.9 (finance-survivor lattice). Write $v_* = (25781, 16266)$ and $v_{1054} = (1054, 665)$. Then

$$25781 \cdot 665 - 1054 \cdot 16266 = 1,$$

so these two vectors are a unimodular basis of $\mathbb{Z}^2$. Every length in $\mathcal{E}_{\text{run}}$, and every one of the 42 packing deaths of Theorem 4.8, is of the form $(L, o_{\min}) = a\, v_* + b\, v_{1054}$. With the table cap $L \leq 10^5$ the 99 survivors fall in three affine slices of counts 29, 47, and 23. The identification with $\mathcal{E}_{\text{run}}$ is Theorem 4.8. This is a change of coordinates for (L, o), not a relation between hypothetical cycles.

Proof. The determinant identity is integer arithmetic. The generator comparison $3^{665} > 2^{1054} > 3^{664}$ gives $o_{\min}(1054) = 665$. Direct evaluation of $o_{\min}$ on the 141 lengths of Theorem 4.6(B) places each pair $(L, o_{\min})$ on the displayed lattice, with the 42 packing deaths the F_1 continuation $b \geq 29$. □

The gap transfer and the short-cycle reduction

Everything above turns a verified descent floor into a per-length exclusion. The floor is the only ingredient that is not scale-free, and along the convergent denominators q_k of $\log 2/\log 3$ the threshold grows like $n_{\max}(q_k) \log n_{\max}(q_k) \asymp q_k q_{k+1}$, equivalently $n_{\max}(q_k) \asymp a_{k+1} q_k^2/\log n$, so every floor leaves the next fan. (The invariant is the first form: over the good convergents $q_k \in \{19, 84, 1054, 50508, 176251\}$ — those with $3^{p_k} > 2^{q_k}$, which are every other one — the ratio $n \log n/(q_k q_{k+1})$ stays in $[0.41, 0.53]$.) This subsection records the one statement of the note that needs no floor. It transfers a lower bound on the linear form

$$\Lambda = \Lambda(L, o) = o \log 3 - L \log 2 = -\log\Big(1 - \theta\Big), \qquad \theta = 1 - \frac{2^L}{3^o},$$

into a bound on the cycle minimum. The only new input is $\log \frac{1}{1-\theta} \le \frac{\theta}{1-\theta}$.

Theorem 4.10 (gap transfer). Let w be a cycle itinerary of length L with o odd letters, based at a cycle minimum $n \ge 2$. Then

$$n \log n \cdot \min(\Lambda, 1) \le 2L.$$

Proof. Write $A = 3^o$, $B = 2^L$, $P = n \log n \ge 0$. Theorem 4.4 reads $P(A - B) \le LA$. If $A \le B$ then $\Lambda \le 0$ and the left side is nonpositive. If $A \ge 2B$, then $A - B \ge A/2$, so $PA/2 \le LA$, hence $P \min(\Lambda, 1) \le P \le 2L$. If $B < A < 2B$, then $\Lambda = \log(A/B) \le A/B - 1 = (A - B)/B$, so $P\Lambda \le P(A - B)/B \le LA/B \le 2L$. □

The Lean form is `cycleMin_gap_transfer`; the abstract corollary "$\varepsilon \le \min(\Lambda, 1)$ implies $n \log n \cdot \varepsilon \le 2L$" is `cycleMin_length_of_gap`.

Corollary 4.11 (short cycles are excluded). Rhin's effective irrationality measure [15], in the packaged form of [12, Lemma 12], gives $\Lambda > \exp(-13.3\,(0.46057 + \log L)) = e^{-6.1256} L^{-13.3}$ for every pair $o < L$ with $3^o \ne 2^L$. Hence every nontrivial cycle satisfies

$$n \log n \;\le\; 2e^{6.1256}\, L^{14.3} \;<\; 915\, L^{14.3}, \qquad \text{equivalently} \qquad L \;>\; \left(\frac{n \log n}{915}\right)^{1/14.3}.$$

In particular a cycle with minimum n and $L^{14.3} \le n \log n/915$ does not exist, for every $n \ge 2$ and without any descent floor.

Proof. On a cycle itinerary $o < L$, so $H = \max(L, o) = L$ in [12, Lemma 12], and $\varepsilon = e^{-6.1256} L^{-13.3} \le 1$. Apply Theorem 4.10 with this ε. □

Remark (what the reduction does and does not do). Corollary 4.11 is a reduction of the no-cycle problem, not a kill. It says that the only cycles left to exclude are the *long* ones, $L > (n \log n/915)^{1/14.3}$; every shorter cycle is excluded by transcendence plus finance alone. At the certified floors of Section 5 this is toothless — at $N_0 = 3.5 \cdot 10^8$ it forces only $L \ge 4$, while the finance table forces $L \ge 780239$ — which is the floor-level statement that a Baker-type transfer cannot compete with the exact gap. The value of the corollary is that it is floor-free and identifies the frontier exactly: the finance survivors of Theorems 4.6 and 5.9 have $L \approx n^{0.59}$ — the measured exponent $\log L/\log n_{\max}(L)$ is 0.595 at $L = 25781$, 0.573 at 50508, 0.582 at 176251 and 0.611 at 780239 — far inside the long regime, and no refinement of the defect *upper* bound can move them into the short one, because along the convergents the required minimum $n_{\max}(q_k)$ grows quadratically in L. Excluding long cycles is a statement about the parity word of a specific orbit at depth L, which no estimate in this note or in the companion discrepancy manuscript [16] controls. The problem "no nontrivial Juggler cycle" is therefore exactly the problem "no long Juggler cycle," and it remains open.

5. The laboratory instance and the walk-charge envelope

Everything in Sections 2–4 is floor-generic: Corollary 4.5 accepts any verified descent floor. This section first records a second, laboratory-certified instance of the same architecture, then replaces the length-only charge by a coupled exponent-walk charge. The extremal walk is identified exactly as a rotation itinerary, and a Denjoy–Koksma bound over certified Ostrowski blocks produces an envelope valid for *every* length in an explicit window — no per-length census and no dynamic program is needed on that window. The kill table at the laboratory floor then yields the period bound 176251, and the second certified floor raises it to 478245 (Corollary 5.10) and the third to 780239 (Corollary 5.11). Throughout, survivors are finance-survivors in the sense of Section 1.

5.1 The laboratory floor

Proposition 5.1 (laboratory descent floor; certified computational input). Every integer $2 \le n \le 26254995$ reaches 1. Precisely: an exact-integer first-passage run records, for each such n, a finite realized itinerary with image strictly below the start; strong induction on that image reaches 1. The run walks the 13127497 odd starts (even starts descend by E) over 106 contiguous chunk records. Three bit-cap seeds (7110201, 13184021, 13782577) are resolved exactly at a $512 \cdot 10^6$-bit cap with exact integer square roots, the largest intermediate having 298912128 bits (seed 7110201); the maximum first passage is 325 steps (seed

15909091). Certificate hashes are in Appendix B. The floor $26254995 = n_{\max}(25781)$ is the cheapest floor that moves the Theorem 4.6 cutoff.

Theorem 5.2 (raised cutoff; verified computation). At $N_0 = 26254995$ the parity table of Theorem 4.6 excludes every length $L \le 50507$: any nontrivial Juggler cycle has period at least 50508. The first finance survivor is $L = 50508$ with $n_{\max}(50508) = 162848324$; 19 parity survivors remain through $L = 2 \cdot 10^5$ (run packing leaves 11 at the same cutoff). This is the one row in the paper where the exact $n_{\max}$ of Section 4 and the value in the committed table differ, and the reason is worth recording: the crossing at $L = 50508$ is sharp to a relative $2.7 \cdot 10^{-10}$, which is *below* the 10^{-9} relative guard the table's generator adds to the right-hand side. That guard is deliberate and conservative — it can only make $n_{\max}$ larger, hence can only refuse to exclude a length — so the table prints 162848325. Both values give the same exclusion here, since the floor 162849448 exceeds either.

Proof. Proposition 5.1 supplies the floor; Corollary 4.5 and the gap table of Proposition 4.4a exclude every L with $n_{\max}(L) \le 26254995$. The contiguous excluded prefix is $L \le 50507$; checksums are Appendix B. □

5.2 Transport to a reduced base

The parity and run-pack tables price valleys independently. On a real minimum-based cycle every state is coupled through one closed exponent walk. With a_k the number of odd letters among the first k, set

$$u_k = (1+\mu)a_k - k, \qquad \mu = \log_2(3/2).$$

The defect-free floors give the upper envelope $x_k \le n^{2^{u_k}}$, and cycle minimality forces $u_k \ge 0$ at every step. The lower envelope requires controlling the accumulated floor losses; that is the transport lemma.

Theorem 5.3 (transport). On a minimum-based cycle with minimum $n \ge 400$, length L, o odd and e even letters, every state satisfies

$$x_k \;\ge\; \left(n\, e^{-D}\right)^{w_k}, \qquad w_k = 2^{u_k}, \qquad D = \frac{1.05\, e}{n} + \frac{0.7\, o}{n^{3/2}}.$$

Proof. Write $\ln x_k \ge w_k \ln n - E_k$. The floor losses give the recursion $E' \le \frac{3}{2}E + 1.05\, x^{-3/2}$ at an odd letter and $E' \le \frac{1}{2}E + 1.05\, x^{-1/2}$ at an even letter, using $-\ln(1-t) \le 1.05\, t$ on $t \le 0.05$. Unrolling, the amplification from injection j to state k is exactly w_k/w_{j+1}. Odd injections have $x_j \ge n$ and $w_{j+1} \ge \frac{3}{2}$, contributing at most $0.7\, n^{-3/2}$ each; even injections have $x_j \ge n^2$ (Theorem 3.2(iii)) and $w_{j+1} \ge 1$, contributing at most $1.05/n$ each. Hence $E_k \le w_k D$. □

The transport inequality is Lean end to end in log form: the walk weight $w_k = 2^{u_k} = 3^{a_k}/2^k$ is rational, so the per-step floor losses (`log_floorPower_odd_ge`, `log_floorPower_even_ge`), the exact weight recursion, and the full induction (`cycleMin_transport`, `WalkTransport.lean`) are elementary real arithmetic over the formalized cycle envelopes `cycleMin_iterate_ge`, `cycleMin_even_ge_sq`, and `cycleMin_prefix_pow_le`.

Consequently the cyclic defect sum $\sum_i 1/(x_i \ln x_i)$ is bounded above by the maximum of the walk charge $\sum_k g(u_k)$, $g(u) = 1/(n'^{2^u} 2^u \ln n')$, over all nonnegative closed exponent walks with o up-steps, evaluated at the *reduced base* $n' = n\, e^{-D}$. No free parameter remains; the walk value feeds the 6/5 unroll of Theorem 4.4 exactly as the parity charge did. At the laboratory floor and window lengths, $D \le 4.6 \cdot 10^{-3}$, so $\ln n' \ge 17.07$.

This consequence is itself Lean end to end (`WalkChargeMax.lean`): writing the charge through the rational weight, $g = 1/(e^{W\nu} W\nu)$ with $W = 2^u = 3^a/2^k$ and $\nu = \ln n'$, exponentiating the transport inequality gives $\sum_k 1/(x_k \ln x_k) \le \sum_k g(w_k)$ (`cycleMin_defect_le_charge`), and the charge of the realized word is dominated by the hug charge of the same length (`cycleMin_defect_le_hug_charge`, using Theorem 5.4 below), all under the recorded hypothesis $\nu > 0$.

5.3 The adversary is the hug itinerary

Theorem 5.4 (hug exchange). Among nonnegative exponent walks with prescribed (L, o), the *hug itinerary* — take E at every step where $u \ge 1$, else O — is prefix-minimal: writing a_k for the odd count of a length-k prefix,

$$a_k^{\text{hug}} \le a_k \qquad \text{for every admissible walk and every } k,$$

equivalently $u_k^{\text{hug}} \le u_k$. Consequently the hug itinerary maximises the walk charge: since g is strictly decreasing in u,

$$g(u_k^{\text{hug}}) \ge g(u_k) \quad \text{for every } k, \qquad \text{hence} \qquad \sum_k g(u_k) \;\le\; \sum_k g(u_k^{\text{hug}}).$$

Proof. At the first disagreement with any other admissible itinerary, the hug itinerary holds E where the other holds O, because hug takes O only when E is illegal and the two words share the same prefix state. The odd-count gap $\delta_k = a_k(\text{other}) - a_k(\text{hug})$ is a path with steps in $\{-1, 0, +1\}$ that cannot go negative, since $\delta = 0$ restores the same state. This is the displayed prefix-minimality $a_k^{\text{hug}} \le a_k$; since $u_k = (1+\mu)a_k - k$ with the same k, it transfers verbatim to $u_k^{\text{hug}} \le u_k$. Feasible pairs ($(1+\mu)o \ge L$) never strand: when the odd budget is exhausted, $u = \text{surplus} + e_{\text{left}} \ge e_{\text{left}}$, so the remaining evens are legal. Applying the strict antitonicity of g termwise and summing over k is the charge comparison. The prefix-minimality core is Lean: `hugOdds_le_of_admissible`. □

Remark (uniqueness). The hug itinerary is in fact the *unique* prefix-minimal admissible path in its (L, o) class, so it uniquely maximises the charge; at the first disagreement the competitor already carries a strictly larger prefix odd count, and strict monotonicity of g makes the total comparison strict. Uniqueness is not used by the kill table; only the displayed domination is.

The analytic half is also Lean, in a strengthened form (`WalkChargeMax.lean`): the charge is antitone in the rational weight (`stateCharge_antitone`, elementary exp monotonicity — no charge integral), so the exact hug itinerary maximises the total charge over *all* admissible exponent walks, not just a fixed (L, o) class (`hug_charge_maximal`). Only the strict within-(L, o) uniqueness of the maximiser remains a human argument.

The statement is about the $u \ge 0$ relaxation, not about realized cycle itineraries. Word-order (Christoffel) prefix-dominance is *false* for this family — the greedy word *OOEO* beats *OOOE* at $(L, o) = (4, 3)$ — so the exchange argument above, not a dominance order, is the correct mechanism.

Realized itineraries do, however, dominate the hug itinerary at the level of odd counts: on any minimum-based cycle itinerary, every length-k prefix carries at least $o_{\min}(k)$ odd letters. This is Lean end to end (`cycleMin_prefix_odds_ge_hug`, `cycleMin_odds_ge_hug`), composing the formalized cycle prefix envelope $2^k \le 3^{a_k}$ (`cycleMin_prefix_pow_le`) with the hug minimality `hugOdds_least`. It is exactly the sense in which the hug itinerary is the cheapest adversary any hypothetical cycle can present. The survivor-lattice generators of Proposition 4.9 lie on this hug diagonal: $(1054, 665)$, $(25781, 16266)$, and the seed $(50508, 31867)$ all satisfy $o = o_{\min}(L)$ (Lean: `hugOdds_1054`, `hugOdds_lattice_base`, `hugOdds_seed`).

Proposition 5.5 (rotation average). The infinite hug walk is the rotation by $\alpha = \log_2(3/2)$ on $\mathbb{R}/(1+\alpha)\mathbb{Z}$. Unique ergodicity of the irrational rotation — extended, in the standard way, from continuous observables to Riemann-integrable ones — gives the charge-per-letter

$$C_*(n') = \frac{1}{\ln 3} \int_1^3 n'^{\,1-t}\, t^{-2}\, dt \;<\; \frac{1}{\ln 3\, \ln n'}.$$

Proof. Substituting $s = (t-1)\ln n'$ gives $C_* = \frac{1}{\ln 3\, \ln n'} \int_0^{2\ln n'} e^{-s}(1 + s/\ln n')^{-2}\, ds$, and the integrand is at most e^{-s}. □

The display and its quantitative sharpening are Lean (`RotationAverage.lean`): the quadratic majorant $t^{-2} \le 1 - 2(t-1) + 3(t-1)^2$ on $[1, 3]$ — the product with t^2 is $1 + 4(t-1)^3 + 3(t-1)^4$ — turns the Laplace bound into an exact antiderivative evaluation, giving $C_*(n') \le (1 - \frac{2}{\nu} + \frac{6}{\nu^2})/(\ln 3\, \nu)$ at $\nu = \ln n'$ with no quadrature (`rotation_average_lt`, `rotationAverage_le`, gap form `rotationAverage_gap`). Only the ergodic *identification* of C_* as the infinite-itinerary average remains classical prose. The observable has one wrap discontinuity, so bare unique ergodicity — uniform Birkhoff convergence for *continuous* observables —

is not invoked directly: the identification uses its standard extension to Riemann-integrable observables of an irrational rotation, and the observable here is monotone with a single jump, hence Riemann integrable and of bounded variation.

This is the infinite-itinerary average, not a finite-L inequality: on the certified survey the finite leftover charge exceeds C_* by up to $1.57 \cdot 10^{-5}$, so $C_L \le C_*$ is false. The next two subsections quantify the finite-L error.

5.4 Itinerary identity

Write C_L for the charge-per-letter of the budgeted hug word at $(L, o_{\min}(L))$.

Lemma 5.6 (itinerary identity). For every L, the budgeted hug itinerary at $(L, o_{\min}(L))$ equals the exact rotation L-prefix generated by the integer rule: E at step k if and only if $3^a \ge 2^{k+1}$, where a is the number of odd letters already used. In particular C_L is a Birkhoff average of the rotation.

Proof. The exact rule keeps $u \in [0, 1+\alpha)$, so its L-prefix uses exactly $o_{\min} = \lceil L \log 2 / \log 3 \rceil$ odd letters. A first budget-forced divergence between the two words would make the exact prefix use more of one letter than its own total, which is impossible. Lean: `budgetedWord_eq_hugWord`, with the window invariant `hugOdds_pow_ge` / `hugOdds_pow_lt` and minimality `hugOdds_least` (`WalkChargeItineraries.lean`). □

5.5 Denjoy–Koksma over certified Ostrowski blocks

Two classical ingredients are used here in a fixed coordinate, so we record both explicitly.

The coordinate change. The infinite hug walk is rotation by $\alpha = \log_2(3/2)$ on the circle $\mathbb{R}/(1+\alpha)\mathbb{Z}$ (Proposition 5.5). Rescaling by $1/(1+\alpha)$ conjugates it to rotation by

$$\theta = \frac{\alpha}{1+\alpha} = \frac{\log(3/2)}{\log 3}$$

on the standard circle $\mathbb{R}/\mathbb{Z}$: with $\alpha = \log(3/2)/\log 2$ one has $1 + \alpha = \log 3/\log 2$, and the quotient is the display. The observable $F(u) = n'^{\,1-2^u}/2^u$ is transported by the same rescaling; a homeomorphic change of coordinate does not change its total variation. All Ostrowski data below — quotients, convergents, digits — refer to this θ.

The Denjoy–Koksma inequality (classical; used as known). If $f : \mathbb{R}/\mathbb{Z} \to \mathbb{R}$ has bounded variation $\mathrm{Var}(f)$ and p_j/q_j is a continued-fraction convergent of the irrational θ, then for every x,

$$\Big|\sum_{k=0}^{q_j-1} f(x + k\theta) - q_j \int_0^1 f\Big| \le \mathrm{Var}(f).$$

The inequality is invoked below only at convergent denominators of θ, never at an arbitrary good rational approximation: the certified pairs (p_j, q_j) are produced by the continued-fraction recurrence from the certified shared quotient prefix of θ (`theta_convergent_denominators`, `theta_convergent_numerators`, `cf_lower_prefix`, `cf_upper_prefix`), so they are genuine convergents of θ, unimodular and coprime (`theta_convergents_unimodular`, `theta_convergents_coprime`). Their approximation quality $|\theta - p_j/q_j| < 1/q_j^2$ and the fact that the q_j rotation steps of one block permute the q_j grid cells are additionally verified in Lean (`theta_convergent_quality`, `theta_block_permutations`); the variation-versus-integral inequality itself is the classical statement and is not re-proved.

Theorem 5.7 (block envelope). For the exact rotation prefix of length L at reduced base n',

$$\big|C_L - C_*(n')\big| \;\le\; \frac{2\,s(L)}{L}, \qquad s(L) = \sum_j b_j,$$

for any decomposition $L = \sum_j b_j q_j$ into convergent denominators q_j of $\theta = \log(3/2)/\log 3$.

Proof. The observable $F(u) = n'^{\,1-2^u}/2^u$ decreases on the circle from $F(0) = 1$ to $F((1+\alpha)^-) = n'^{-2}/3$, so its variation including the wrap jump is < 2; the rescaled observable on $\mathbb{R}/\mathbb{Z}$ has the same variation. The Denjoy–Koksma inequality stated above, applied per block with the certified convergent quality $|\theta - p_j/q_j| < 1/q_j^2$, bounds the ergodic sum of each block of length q_j within $\mathrm{Var}(F)$ of $q_j C_*$. The decomposition $L = \sum_j b_j q_j$ partitions the length-L trajectory segment into consecutive blocks whose starting phases differ; since Denjoy–Koksma holds uniformly in the starting phase x, it applies independently to each block, whatever phase that block inherits from its predecessor. Summing the $s(L)$ blocks gives the display. $\square$

The denominator list

$$1,\ 2,\ 3,\ 8,\ 19,\ 65,\ 84,\ 485,\ 1054,\ 24727,\ 50508,\ 125743,\ 176251$$

is certified by an interval continued fraction on the big-integer sandwich $2^{17087915} > 3^{10781274}$ and $2^{16785921} < 3^{10590737}$. The sandwich, the resulting real bounds on θ, the shared quotient prefix of the two rational endpoints, and the convergent recurrence are Lean (`theta_sandwich_upper`, `theta_sandwich_lower`, `lower_lt_walkTheta`, `walkTheta_lt_upper`, `cf_lower_prefix`, `cf_upper_prefix`, `theta_convergent_denominators`, `OstrowskiSandwich.lean`); the cylinder-interval bridge from the endpoints to θ itself is classical. The quantitative hypothesis Denjoy–Koksma needs per block is also Lean: with the matching numerators $0, 1, 1, 3, 7, 24, 31, 179, 389, 9126, 18641, 46408, 65049$ (`theta_convergent_numerators`), consecutive pairs are unimodular, all pairs are coprime, every certified convergent satisfies $|\theta - p/q| < 1/q^2$ against the sandwich (`theta_convergents_unimodular`, `theta_convergents_coprime`, `theta_convergent_quality`), and each block's q rotation steps permute the q grid cells (`theta_block_permutations`). Only the classical variation-versus-integral inequality itself remains prose. A Koksma-type bound with constant 1 (that is, $+1/L$) is *false* for this observable; the correct constant is $2s(L)$.

5.6 The window theorem

Theorem 5.8 (uniform window envelope). For every $L \in [50508,\ 16785921)$, at the laboratory floor,

$$C_L \ \le\ C_*(n') + \frac{2\,s(L)}{L}\ <\ \frac{1}{\ln 3\ \ln n'}.$$

Proof. Greedy Ostrowski digits obey $b_j \le a_{j+1}$, so with the certified quotients $\theta = [0; 2, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, 1, 55, ...]$ the digit sum satisfies $s(L) \le \sum_{j\le 13} a_j = 47$ for $L < q_{13} = 301994$, and on the remaining range $L = b\,q_{13} + r$ with $1 \le b \le a_{14} = 55$ and $r < q_{13}$, so $s(L) \le b + 47$. This step is Lean: the digit cap, exact reconstruction $L = \sum_j b_j q_j$, and the digit-sum bound hold for *any* denominator sequence satisfying the convergent recurrence (`ostroDigit_le`, `ostro_sum_eq`, `ostro_digitSum_le`, `OstrowskiNumeration.lean`), and the certified θ instance gives the two caps structurally (`theta_digitSum_le`, `greedyDigitSum_le`); the sandwich already reaches q_{14} and q_{15} (`theta_sandwich_lower`, `theta_sandwich_upper`, $2^{16785921} < 3^{10590737}$ and $3^{10781274} < 2^{17087915}$). The transport deficit keeps $\ln n' \ge 17.07$, hence

$$\frac{2\,s(L)}{L}\ \le\ 9.38\cdot 10^{-4}\ <\ 5.14\cdot 10^{-3}\ \le\ \frac{1}{\ln 3\ \ln n'} - C_*(n'),$$

the last gap from $\int_0^{2\ln n} e^{-s}(1 + s/\ln n)^{-2} ds \le 1 - 2/\ln n + 6/(\ln n)^2$, which is Lean (`rotation_average_le`, gap form `rotationAverage_gap`, `RotationAverage.lean`). Conclude by Lemma 5.6 and Theorem 5.7. $\square$

Why the window reaches q_{14}, and why that is the natural stop. The bound $2s(L)/L$ is worst at the *small* end, not the large one: a large Ostrowski digit forces a large L, since $b_j = c$ requires $L \ge c\,q_j$. On $[q_{13}, q_{14})$ the two effects give $2s(L)/L \le 2(b+47)/(b\,q_{13}) \le 3.18\cdot 10^{-4}$, maximised at $b = 1$ and an order below the value at $L = 50508$; an exhaustive scan of $[50508, 2\cdot 10^6)$ puts the true maximum at $9.3766\cdot 10^{-4}$, attained at $L = 74654$ with $s = 35$. So the window costs nothing to extend across the whole of the $a_{14} = 55$ fan, and it stops at $q_{14} = 16785921$ only because that is where the next partial quotient begins. The right-hand side is what shrinks: writing the gap in closed form,

$$\frac{1}{\ln 3\ \ln n'} - C_*(n') = \frac{2\ln n' - 6}{\ln 3\,(\ln n')^3},$$

which is $5.14 \cdot 10^{-3}$ at $\ln n' = 17.07$ and falls to $3.82 \cdot 10^{-3}$ at the floor of Corollary 5.14. Against the window maximum $9.38 \cdot 10^{-4}$ that leaves a factor between 4.1 and 5.5 at every floor in this paper, and the window theorem survives floors up to $n' \approx 2.8 \cdot 10^{18}$.

Two consequences, and one thing that is *not* a consequence. First, $q_{14} = 16785921$ is exactly L_{55}, the last member of the semiconvergent fan of Proposition 5.12, so the window now covers the entire fan. Second, the window theorem is not what limits the walk charge; that is Remark 5.8a.

What does *not* follow is that the kill tables become census-free. The window theorem is a uniform upper bound on the **charge**, so no per-length dynamic program is needed to bound $B(L)$ anywhere on $[50508, 16785921)$. The kill decision is the comparison $\theta(L) > \frac{6}{5}B(L) \cdot \text{guard}$, and its left-hand side $\theta(L) = 1 - 2^L/3^{o_{\min}(L)}$ is a per-length Diophantine quantity that the envelope says nothing about. Corollaries 5.10 and 5.11 identify exactly that quantity as the obstruction at the surviving fan members. So the extension removes the per-length *dynamic program* from the whole fan and leaves the per-length *arithmetic* in place. Eliminating the latter would need a lower bound on the Diophantine deficit that is uniform along the fan — a genuinely different statement, and one this paper does not prove.

Remark 5.8a (what the walk charge can ever buy). The charge of Theorem 5.3 prices a state at exponent u by $f(u) = 1/(x \ln x)$ with $x = (n')^{2^u}$, so $f(u)/f(0) = 2^{-u} \exp(-(2^u - 1) \ln n')$. Since $2^u - 1 = u \ln 2 + O(u^2)$, the charge decays on the scale $u \asymp 1/\ln n'$: it is a Laplace-type boundary layer at $u = 0$, not a hard cutoff, and $1/(\ln 3 \ln n')$ — the same quantity Theorem 5.8 carries — is the scale of its effective mass rather than the width of an interval. Integrating the layer against the walk gives an *empirical scaling law* for the advantage over the length-only parity charge,

$$\frac{\text{parity}}{\text{walk}} \approx c \ln n', \qquad c \approx 0.44.$$

The constant is fitted, not derived; only the $\ln n'$ dependence is explained by the boundary layer. Measured at $L = 50508$ over ten orders of magnitude in the floor, the ratio (improvement)$/ \ln n'$ runs 0.461, 0.451, 0.446, 0.445, 0.439, 0.432, 0.427 at $n_0 = 10^6, 2.6 \cdot 10^7, 1.6 \cdot 10^8, 3.5 \cdot 10^8, 10^{10}, 10^{13}, 10^{16}$: constant to 8% across the range.

So the walk charge is neither exhausted nor cheap to improve. It is not limited by certification depth, which the extension above showed has room to spare, and it is not limited by the envelope: substituting the census-free envelope of Theorem 5.8 for the exact lattice program changes the margin at $L = 50508$ from 1.1204 to 1.1196, a difference of 0.07%. It is limited by the shape of f. Since the advantage grows like $\ln n'$, **doubling the walk charge's efficiency requires squaring the descent floor**.

One caution about that cross-check, because it is easy to over-read. Both quantities compared there live on the *relaxed* class: the lattice program admits every binary word with o odds, e evens and $u_k \geq 0$, realizable or not. The agreement to 0.07% therefore says the envelope is tight against the relaxed optimum, and says nothing about how far that optimum sits above the true maximum over *realizable* words. That is a separate question, and it is the only place in this construction where a factor of the size in question could plausibly hide, so it is worth measuring rather than assuming.

Proposition 5.15 (the relaxation cannot hide a constant). For $14 \leq L \leq 24$ and $o = o_{\min}(L)$, let $\mathcal{A}(L)$ be the class the lattice program maximises over — all masks with o odds whose exponent walk stays nonnegative — and let $\mathcal{R}(L) \subseteq \mathcal{A}(L)$ be those realized as $\text{word}_L(m)$ for some odd $m < 2 \cdot 10^7$. Then

1. the charge-maximising element of $\mathcal{A}(L)$ lies in $\mathcal{R}(L)$ at every one of those eleven lengths, so the two maxima agree exactly; and
2. that agreement does not depend on (1). Ordering $\mathcal{A}(L)$ by charge, the second element carries at least 0.9846 of the maximum and the tenth at least 0.9214, both ratios *increasing* in L (at $L = 24$: 0.9898 and 0.9481); and membership of $\mathcal{R}(L)$ is independent of charge to measurement accuracy — at $L = 24$ the realized fraction is $0.698, 0.702, 0.689, 0.694, 0.689, 0.691, 0.697, 0.698, 0.697, 0.695$ across the ten charge deciles. So the first realized word appears within the first few ranks whatever the argmax does, and the realized maximum is within about one percent of the relaxed maximum regardless.

Discussion. The relaxation is not vacuous: the realized fraction $|\mathcal{R}|/|\mathcal{A}|$ is 1.000 through $L = 21$ and then falls to 0.992, 0.906, 0.695, so by $L = 24$ the program maximises over half again as many words as can occur. But the words it adds are spread uniformly through the charge ordering rather than concentrated where the charge is small, and the ordering is flat at the top; those two facts together, not the coincidence in (1), are what bound the slack.

It is worth recording what this rules out. A relaxation that cost a factor of 8 would have to remove almost the entire top of the charge ordering, and $\mathcal{R}$ removes about 30% of it uniformly. Whatever explains the walk charge's advantage being $\approx 0.44 \ln n'$ rather than something larger, it is not the exponent-walk relaxation.

Proposition 5.16 (flatness at the kill-table lengths). Part (2) of Proposition 5.15 rests on the charge ordering being flat at the top, and enumeration establishes that only for $L \leq 24$. It can be established directly at the lengths the kill tables use, without enumerating, by running the lattice program of Theorem 5.3 with the K best partial sums per state in place of the best one: max-plus becomes top-K-plus, the rolling array grows by a factor K, and the pass stays linear in L. At $K = 16$, writing r_j for the j-th largest charge,

$$1 - \frac{r_{16}}{r_1} = \begin{cases} 9.44 \cdot 10^{-2}, & L = 18, \\ 6.24 \cdot 10^{-2}, & L = 24, \\ 5.42 \cdot 10^{-8}, & L = 50508,\ n' = 2.6 \cdot 10^{7}, \\ 6.80 \cdot 10^{-9}, & L = 176251,\ n' = 1.6 \cdot 10^{8}, \\ 1.08 \cdot 10^{-9}, & L = 780239,\ n' = 3.5 \cdot 10^{8}. \end{cases}$$

So the top does not merely stay flat at the operative lengths: it flattens by seven orders of magnitude between $L = 24$ and $L = 50508$, and continues to flatten with L.

Proof of the computation. The recursion and admissible class are those of Theorem 5.3; only the accumulator changes. The top-K program reproduces the exhaustive top ten at $L = 18$ to 10^{-18}, and its GPU form agrees with the host form bitwise, so the values above are the same objects the kill tables use. $\square$

The reason is visible in the charge. At the operative lengths the sum is carried by the $\asymp L/(\ln 3 \ln n')$ states with $u \approx 0$, and the closest a nonnegative walk can come to $u = 0$ other than exactly is $\asymp 1/o$; perturbing the extremal walk therefore moves one contribution by a relative $\asymp \ln n'/o$ out of a total of $\asymp L/\ln n'$ equal terms. At $L = 50508$ that predicts $\approx 8 \cdot 10^{-8}$ against the measured $5.4 \cdot 10^{-8}$.

Together with Proposition 5.15 this closes the question the relaxation raised. Whatever removes 30% of the admissible words, it must remove *all sixteen* of the leading walks before the realized maximum falls by as much as $5 \cdot 10^{-8}$; realizability is spread uniformly through the ordering, so that is not what is happening. The exponent-walk relaxation is not hiding a constant, at the lengths where the constant would matter.

One limit remains on how far this should be read. $\mathcal{R}(L)$ is a scanned lower bound on the realizable set, so a word missing from it can only raise the realized maximum: the measurement bounds the slack above, which is the direction that matters. An earlier and thinner scan ($m < 2 \cdot 10^{6}$) put the $L = 22$ argmax outside $\mathcal{R}$ and the realized fraction at 0.38; at $2 \cdot 10^{7}$ both reverse, which is the expected behaviour when a length-L word occurs with density $\approx 2^{-L}$, and a caution against reading one absence as an exclusion.

What the boundary layer does suggest is that improvement will not come from a better envelope for this charge. It raises a sharper question than "find a better charge": is *every* charge that depends only on the transported exponent u, under the same one-step defect budget, subject to the same $O(1/\ln n')$ concentration? A positive answer would turn the observation above into a limitation theorem for the whole class, and would be worth more than another constant.

5.7 Kill table and the period bound

The envelope controls the charge; kill decisions are the per-length finance comparison $\theta(L) > \frac{6}{5}\, B(L) \cdot \mathrm{guard}$ of the Theorem 4.6 architecture, which is Diophantine, not envelope-limited.

The comparison template is itself Lean (`cycleMin_hug_kill_criterion`, `DefectFinance.lean`): every minimum-based cycle at $n \geq 400$ with positive reduced log-base must satisfy $1 - 2^{L}/3^{o} \leq \frac{6}{5} \sum_k g(\mathrm{hugWeight}_k)$

at the reduced base, chaining the Lean finance inequality (`cycleMin_defect_finance`, the certified identity of Theorem 4.6) with the defect-to-hug-charge envelope of §5.2. The kill decisions below verify the numeric *failure* of this inequality per length; only that evaluation, and the parity refinement of Corollary 4.5, remain verified computation.

Theorem 5.9 (verified computation; period bound at the laboratory floor). At $N_0 = 26254995$, the walk charge of Theorem 5.3 excludes 18 of the 19 parity survivors of Theorem 5.2 through $L = 2 \cdot 10^5$. Kill margins run from 1.1204 at the seed $L = 50508$ (required improvement over parity 6.87, walk supplies 7.70; deficit $D = 7.4566 \cdot 10^{-4}$), through 1.1195 and 1.1187 at its multiples 101016 and 151524, up to 7.69 at the parity-marginal lengths. The sole walk survivor is $L = 176251$ (margin 0.159; required improvement 48). The combined parity + walk contiguous excluded prefix is 176250: any nontrivial Juggler cycle has period at least 176251.

Proof. Theorem 5.2 leaves the 19 parity survivors. Each is priced by the exact (step, odd-count) lattice dynamic program at the reduced base (Theorem 5.3) and compared against $\theta(L)$ under the 6/5 unroll; substituting the census-free envelope of Theorem 5.8 instead recovers the same 18 kills (margin 1.1196 at $L = 50508$; $L = 176251$ survives at 0.1588). Checksums are Appendix B. □

Calibration. At the base instance $(L, N_0) = (25781, 10^6)$ the walk charge gives margin 0.196 — correctly no kill (required improvement 32.5, walk supplies 6.37). The mechanism did not change between the two floors; the target did.

Corollary 5.10 (verified computation; second laboratory floor). Every integer $2 \leq n \leq 162849448$ reaches 1 (first-passage certificate exactly as in Proposition 5.1: the extension segment walks 68297226 odd starts over 547 contiguous chunk records with exact integer square roots and no failures of any kind; the maximum first passage is 433 steps at seed 78641579, and the largest intermediate has 463362780 bits at seed 92502777; hashes in Appendix B). At $N_0 = 162849448$ the parity table excludes the previous survivor cluster $\{50508, 101016, 151524\}$ outright and leaves 25 survivors through $L = 6 \cdot 10^5$; the walk charge of Theorem 5.3 kills the 15 below 478245 (margins 1.198 at 176251 and 352502, up to 8.44); lengths $L \leq 176250$ stay excluded because both exclusions are monotone in the floor. Every survivor here lies inside the window $[50508, 16785921)$ of Theorem 5.8, so the *charge* side of each kill is census-free; the per-length lattice program is run as a cross-check and agrees to 0.07%. The comparison against $\theta(L)$ remains per-length. The combined contiguous excluded prefix is 478244: any nontrivial Juggler cycle has period at least 478245.

The sole survivor at this floor is the semiconvergent fan member $478245 = 176251 + 301994$: required improvement over parity 19.46, direct walk margin 0.4334 — the obstruction is the Diophantine quality of $|3^o - 2^L|$ along the fan, not the envelope. The kill table recomputes on commodity hardware in minutes (Appendix B). Corollary 5.11 evaluates the same criterion at the next cheap floor.

Corollary 5.11 (verified computation; third laboratory floor). Every integer $2 \leq n \leq 350000000$ reaches 1 (first-passage certificate exactly as in Proposition 5.1: the extension segment from 162849449 walks 93575276 odd starts over 749 contiguous chunk records with exact integer square roots; two bit-cap seeds 172376627 and 240154767 are resolved at a $3 \cdot 10^9$-bit cap, the largest intermediate having 1493770145 bits at seed 172376627; the maximum first passage is 466 steps at seed 198424189; hashes in Appendix B). At $N_0 = 350000000$ the parity table leaves 17 survivors through $L = 8 \cdot 10^5$; the five leftovers below 478245 stay excluded because both exclusions are monotone in the floor, and the walk charge of Theorem 5.3 kills the 10 leftovers in $[478245, 755512]$ (margins 1.00555 at 478245, up to 7.824 at 504026). Every survivor here lies inside the window $[50508, 16785921)$ of Theorem 5.8, so the *charge* side of each kill is census-free and the per-length lattice program is only a cross-check; the comparison against $\theta(L)$ remains per-length. The combined contiguous excluded prefix is 780238: any nontrivial Juggler cycle has period at least 780239. This is the main numerical result of the paper.

The sole survivor is the semiconvergent fan member $780239 = 176251 + 2 \cdot 301994$: required improvement over parity 14.46, direct walk margin 0.6049 — the obstruction remains the Diophantine quality of $|3^o - 2^L|$ along the fan, not the envelope. The kill table is the GPU floating-point certified comparison of the same Theorem 5.9 criterion (Appendix B).

Beyond $q_{13} = 301994$ the window theorem needs deeper certified quotients of θ, and killing the remaining near-convergent survivors (starting with the fan member $L = 780239$) is a Diophantine question about $|3^o - 2^L|$; neither is attempted here. The family leftover — the semiconvergent fans of $\log 2/\log 3$ reduced to dangerous-position partial quotients — is the working draft juggler_near_convergent_diophantine_note.md. The survivors are finance-survivors, not candidate cycles.

5.8 The price of the fan

The paragraph above says the obstruction is Diophantine and stops. It can be made quantitative, and the answer is short: the frontier lengths form one explicit arithmetic progression of 56 terms, ending on the next convergent, and the descent floor each term costs is computable in closed form. Nothing here is a new exclusion. It is a price list for the exclusions that remain, and it says exactly what the walk charge of this section bought.

Proposition 5.12 (fan law). Let $q_{12} = 176251$, $q_{13} = 301994$ be the consecutive convergent denominators of $\log 2/\log 3$ bracketing the frontier, with numerators $p_{12} = 111202$, $p_{13} = 190537$. Put

$$L_k = q_{12} + k\, q_{13}, \qquad o_k = p_{12} + k\, p_{13}, \qquad 0 \le k \le 55.$$

Then $o_k = o_{\min}(L_k)$ for every such k; the linear form is exactly affine in k,

$$\Lambda_k = o_k \log 3 - L_k \log 2 = \Lambda_0 + k\Lambda', \qquad \Lambda_0 = 3.6002 \cdot 10^{-6}, \quad \Lambda' = -6.4508 \cdot 10^{-8};$$

and $k = 55$ is the last index with $\Lambda_k > 0$, because $\Lambda_0/|\Lambda'| = 55.81$. At that last index $L_{55} = 16785921 = q_{14}$ and $o_{55} = 10590737 = p_{14}$: the fan ends on the next convergent, which is where the partial quotient $a_{14} = 55$ is consumed.

Proof. Λ is linear in (o, L) and $(o_k, L_k) = (p_{12}, q_{12}) + k\,(p_{13}, q_{13})$, which is the affine formula; $\Lambda' = \Lambda(q_{13}) < 0$ because $3^{p_{13}} < 2^{q_{13}}$, consecutive convergents lying on opposite sides. Since $\Lambda_k \in (0, \log 3)$ for $0 \le k \le 55$, one has $3^{o_k} > 2^{L_k} > 3^{o_k - 1}$, which is $o_k = o_{\min}(L_k)$. The sign change at 55.81 is arithmetic, and $q_{14} = a_{14} q_{13} + q_{12}$ with $a_{14} = 55$ identifies the endpoint. ☐

Because Λ_k *decreases* in k, so does $\theta(L_k)$, and $n_{\max}(L_k)$ increases: the fan grows strictly more expensive as it is climbed, and the cheapest member is always the one at the current frontier. The resulting rule is a single comparison. For every $k \ge 1$,

$$N_0 \;\ge\; n_{\max}(L_k) \qquad \Longrightarrow \qquad \text{period} \;\ge\; L_{k+1},$$

verified directly at $k = 1, \ldots, 12$ and at $k = 20, 31, 40, 52, 54$. (At $k = 0$ alone one extra length intervenes, the doubling $2q_{12} = 352502$, whose threshold 1044095006 sits 1793 above $n_{\max}(q_{12})$; the multiples mq_{12} cluster just above $n_{\max}(q_{12})$ in the same way and are all cleared by $n_{\max}(L_1)$.)

k	L_k	Λ_k	$n_{\max}(L_k)$ = floor that passes L_k
0	176251	$3.600 \cdot 10^{-6}$	$1.044 \cdot 10^{9}$
1	478245	$3.536 \cdot 10^{-6}$	$2.756 \cdot 10^{9}$
2	780239	$3.471 \cdot 10^{-6}$	$4.480 \cdot 10^{9}$
3	1082233	$3.407 \cdot 10^{-6}$	$6.238 \cdot 10^{9}$
6	1988215	$3.213 \cdot 10^{-6}$	$1.182 \cdot 10^{10}$
31	9538065	$1.600 \cdot 10^{-6}$	$1.040 \cdot 10^{11}$
52	15879939	$0.246 \cdot 10^{-6}$	$1.034 \cdot 10^{12}$
54	16483927	$0.117 \cdot 10^{-6}$	$2.199 \cdot 10^{12}$
55	16785921	$0.052 \cdot 10^{-6}$	$4.866 \cdot 10^{12}$

Three readings of this table.

What the walk charge is worth. Corollary 5.11 reaches $L_2 = 780239$ from $N_0 = 3.5 \cdot 10^{8}$. Finance and parity alone reach it only from $n_{\max}(L_1) = 2.756 \cdot 10^{9}$. The walk charge is therefore worth a factor 7.9 in descent floor at the present frontier — and a factor 6.4 at the previous one, where Corollary 5.10 reached L_1 from $1.63 \cdot 10^8$ against a finance requirement of $1.044 \cdot 10^9$.

What the next step costs. A purely computational route to $L_3 = 1082233$ — no walk charge, no new idea, only a longer first-passage run — needs $N_0 \geq n_{\max}(780239) = 4479642886$, a floor 12.8 times the present one. The walk charge does very much better, and by a factor that can be measured rather than guessed.

Lemma 5.13 (margin scaling). At fixed L, the certified kill margin $\theta(L)/(\frac{6}{5}B(L, N_0))$ of Theorem 5.9 grows like $(N_0 \log N_0)^\beta$ with $\beta = 1.047$.

Evidence. The committed kill records contain two lengths priced at two floors each. At $L = 176251$ the margin runs 0.158796 at $N_0 = 26254995$ to 1.198309 at 162849448; at $L = 478245$ it runs 0.433383 at 162849448 to 1.005552 at 350000000. The implied exponents are 1.0491 and 1.0458, agreeing to 0.31%. The excess over $\beta = 1$ is the secondary dependence of the reduced base on the floor; it is not needed for the conclusion below, which is unchanged at $\beta = 1$.

Applied to the present survivor, $L = 780239$ at margin 0.604888, this predicts a kill at $N_0 = 5.53 \cdot 10^8$. Evaluating the criterion directly gives

$$\text{the walk charge kills } L = 780239 \text{ from } N_0 = 553906250,$$

so the prediction is accurate to 0.2%, and the walk charge is worth a factor $4479642886/553906250 = 8.09$ in descent floor here — against 7.9 at the previous frontier and 6.4 at the one before. The efficiency is stable, and the 56-step price list above may be read divided by roughly 8 whenever the walk charge is applied.

Corollary 5.14 (conditional; the next period bound). If every integer $2 \leq n \leq 554000000$ reaches 1, then any nontrivial Juggler cycle has period at least 1082233.

Proof, modulo the floor. At $N_0 = 554000000$ the parity table of Corollary 4.5 leaves twenty survivors below $L_3 = 1082233$, and the walk charge of Theorem 5.3 kills all of them. Everything at or below 780239 is excluded because both exclusions are monotone in the floor and $554000000 > 350000000$, except 780239 itself, which is killed here at margin 1.000884. The nine survivors above it are killed with margins

806020	8.076	830747	2.908	881255	4.596
931763	6.101	956490	1.663	982271	7.451
1006998	3.203	1032779	8.669	1057506	4.595

all by the certified evaluation of the Theorem 5.9 comparison at the reduced base. The next length, $L_3 =$ 1082233, survives at margin 0.70815. Hence the contiguous excluded prefix is 1082232. □

The kill table is therefore *already done*; the only missing input is the descent floor. Extending the certified floor from $3.5 \cdot 10^8$ to $5.54 \cdot 10^8$ is a first-passage run over roughly $1.0 \cdot 10^8$ further odd starts — comparable in size to the extension Corollary 5.11 already performed, and a factor 1.58 in floor rather than the 12.8 that finance alone would demand. We do not carry out that run here; Corollary 5.14 is stated conditionally so that the computation and the criterion are separable, which is the architecture of Corollaries 5.10 and 5.11 as well.

Where computation ends. Exhausting the fan, i.e. reaching $L_{55} = 16785921$, needs $N_0 \geq 2.20 \cdot 10^{12}$ by finance and about $2.7 \cdot 10^{11}$ with the walk charge; passing the convergent itself needs $4.87 \cdot 10^{12}$, respectively $6.0 \cdot 10^{11}$, after which the same structure repeats one scale up with q_{14} in the role of q_{12}, at a cost that grows like $a_{15} q_{14}^2 / \log n$.

So the bound is not stuck against a wall; it is on a staircase whose steps are priced — whose next step costs a factor 1.58 in floor with the walk charge (12.8 without it), and whose 56 steps together cost a factor 4660 either way, since the walk charge divides the whole list by a constant. What no floor buys is the end of the staircase. The fans recur at every convergent, the required floor grows quadratically in the length, and a period bound obtained this way can never become a proof that no cycle exists. That gap is not computational, and it is the subject of the remark after Corollary 4.11.

6. Limitations and future directions

The result does not imply termination. The remaining finance-survivor lengths are uncontrolled, and existence of a cycle at such a length is open.

A start $n \geq 2$ has a *descent certificate* if there exists a realized finite itinerary w with $J^{|w|}(n) < n$. Even starts realize E; an odd start with even image realizes OE. The complement of those two words is the odd-to-odd class. If every start above 1 has some descent certificate, strong induction yields arrival at 1. That hypothesis is not proved, and a uniform run bound on expanding blocks is unavailable: four consecutive expanding blocks occur already at

$$1999 \xrightarrow{OOE} 5169 \xrightarrow{OOOOEE} 50093 \xrightarrow{OOE} 193753 \xrightarrow{OOE} 887471.$$

The number p of odd runs on a minimum-based cycle — equivalently, the number of excursions on the necklace of Section 4 — looks like the next concrete direction. The run form gives $p \leq e$ and, because the first odd run has length at least two, $p \leq o-1$, hence $p \leq \min(e, o-1) < 0.3691\,L$ on an expanding itinerary; that is only the trivial ceiling, and a genuine lower bound on p, or a peak-height / peak-count tradeoff, would feed Theorem 4.7.

It is worth recording why we do not pursue it. A bound on p is only useful if it constrains the *adversary*, and for the walk charge the adversary is the extremal walk of Theorem 5.3, which can be recovered from the lattice program by storing its decisions. That walk turns out to saturate both ceilings at once. Writing p for its odd-run count:

L	84	1054	25781	50508
p	31	389	9515	18641
$\min(e, o-1)$	31	389	9515	18641
longest odd run	2	2	2	2

So $p = e$ exactly, at every length tested: every even run has length one, and the walk is a word in the two blocks `OE` and `OOE` alone, mixed at the density $\log 2/\log 3$ — the hug itinerary of Theorem 5.4, in its Sturmian form. At $L = 84$ it is $\mathtt{O}^2\mathtt{E}\,\mathtt{O}^2\mathtt{E}\,\mathtt{O}^2\mathtt{E}\,\mathtt{OE}\,\mathtt{O}^2\mathtt{E}\,\mathtt{O}^2\mathtt{E}\,\mathtt{OE}\cdots$.

Both halves of the direction are therefore closed against this adversary. A *lower* bound on p cannot bite, because p is already at its combinatorial maximum. A peak-height / peak-count tradeoff cannot bite either, because the longest odd run is 2 and the peaks are already as low as an expanding word allows. The extremal walk is the flattest word available, and it is flat in both senses at once.

One further check falls out. That walk begins `OO` and ends `E`, so it satisfies the minimum-based structural restrictions of Theorem 3.2: Section 3 does not cut the adversary down either, which is the qualitative reason behind the measurement in Proposition 5.15. What would bite is a constraint that forbids the `OE`/`OOE` mixture itself at the critical density — an arithmetic statement about which Sturmian words are realizable as Juggler itineraries, not a counting statement about runs. That statement is false as stated, and it is worth saying why.

The hug word is realized at the generic rate. Counting odd $m < 4 \cdot 10^6$ whose itinerary begins with the length-l prefix of the extremal walk, against the $2^{-(l-1)}$ a generic word would get: the ratio is 1.00 at $l = 8, 9, 10$ and drifts to 1.38 by $l = 18$. The control is the same statistic over every itinerary that occurs at depth 18 — 34342 of them — whose ratios have median 1.05 and run from 0.066 to $4.4 \cdot 10^3$. The hug word therefore sits just above the median and deep inside the bulk: by realizability it is an ordinary word, not a rare one.

So the walk charge is not bounding an adversary that cannot occur. That is a point in the construction's favour, and it also closes the reformulation: the obstruction is not the realizability of the extremal word. Taken with Propositions 5.15 and 5.16, and with Remark 5.8a, every candidate explanation for the walk charge's $0.44 \ln n'$ has now been eliminated — the envelope (tight to 0.07%), the certification depth (a factor 55 of unused range), the exponent-walk relaxation (a part in 10^8 at the operative lengths), the run count, and the realizability of the adversary. What remains is the shape of the charge itself, and that is not a slack to be recovered but the value of the method. The remaining gap recorded there is the missing link from the forced lift at the minimum, through the complete necklace, to the entry one-step preimage; it is not a halt theorem.

Corollary 4.11 fixes what a full exclusion would have to prove. Every method of this note bounds the same side of one equation, $\Lambda = \sum_i \delta_i/\log x_i$, and along the convergents of $\log 2/\log 3$ the minimum required of a survivor grows quadratically in the period, so no descent floor and no refinement of the defect upper bound can exclude all lengths. Transcendence excludes the short cycles $L^{14.3} \le n \log n/915$ unconditionally. What remains is the long regime, and there the obstruction would have to come from the parity word of a specific orbit at depth L — for a prescribed word w the integers m with $f_w(m) = m$ under the prescribed branches number about $1/(\Lambda(1+\log n)) \approx n/L$ and sit in a band around the finance balance point, and a cycle exists exactly when one of them realizes w. On the hug words at $L = 19, 84, 1054$ the band holds 11, 55, 1689 integers and the realized parity depth on it is a fair coin (mean 1.03 at $L = 1054$, maximum $8 < \log_2 1689$). No estimate here, and none in the companion discrepancy manuscript [16] (whose per-depth control is complete through depth four and covers two words of depth five, and is averaged over starts), addresses a single orbit at that depth. We record this as the open problem, not as a program.

The same pattern — a piecewise power map, integer rounding, and a cycle minimum — produces a defect-financing obstruction. The Juggler-specific content is the interaction of $x^{3/2}$ and $x^{1/2}$. Analogous questions for other piecewise floor-power maps are not taken up here.

6.1 What the companion manuscripts add, and what they do not

The results of [16] and [17] postdate the theorems above. They change the *context* of the cycle problem in five ways, none of which excludes a cycle.

The envelope becomes a descent step. Theorem 2.2 bounds the state after an itinerary w by $J^{|w|}(n)^{2^{|w|}} \le n^{3^{\#O(w)}}$. In [17] this is used in the other direction: if the exponent walk $u_t = o_t \log_2 3 - t$ of a start $n \in (y, 2y]$ reaches $-L(y)$, $L(y) = \log_2(\log 2y/\log N_0)$, within t steps, then $J^t(n) \le N_0$, so n reaches 1 by the certified floor. The floor of Section 5 is thereby the *target* of a Tao-type reduction: a bound $\#\{n \text{ odd} \in (y, 2y] : J^t(n) > N_0 \ \forall t \le C \log_2 \log y\} \le y(\log y)^{-e}$ with $e > 0.595$ implies the whole conjecture, cycles included ([17], Theorems 3 and 4). A larger floor lowers $L(y)$: the certified $3.5 \cdot 10^8$ against the Lean-verified 260 is worth $\log_2(19.67/5.56) = 1.82$ units of the walk, which lowers the required depth $C\,L(y)$ by 35–38 letters ($C = 19$ to 21) at every scale y. That is all the floor does for the asymptotics; it crosses no threshold.

Cycle basins are contagious. If a nontrivial cycle C exists, its basin $B(C) = \{n : \exists k,\ J^k(n) \in C\}$ is backward-closed, and [17, Theorem 1] gives $\sum_{n \in B(C),\, n \le x} 1/n \gg (\log x)^\lambda$ for every $\lambda < 0.4050$: on infinitely many dyadic blocks the starts that enter C have natural density $\gg (\log y)^{-0.6}$. The two constraints do not meet. This paper bounds the *states* of C — minimum above $3.5 \cdot 10^8$, period at least 780239, at least four even steps — and thereby the seed of the basin ($\sum_{x \in C} 1/x \le L/\min C$); contagion bounds the growth of the basin from any seed, from below; no inequality in either paper bounds a basin from above. A cycle would be rare in its states and common in its basin, and neither statement contradicts the other.

Cycles sit at the critical odd share. The gap transfer of Theorem 4.10 forces $0 < \Lambda = o \log 3 - L \log 2 \le 2L/(n \log n)$ on a cycle with minimum $n > 2L/\log n$, so $o/L = \log 2/\log 3 + \Lambda/(L \log 3)$ with $\Lambda/L \le 2/(n \log n)$. The weakest hypothesis of the reduction in [17] — the no-momentum form — asks that the tilted odd share of the starts still above the floor stay below some $q < \log 2/\log 3 = 0.6309$, on average over depths; a cycle's word is a periodic itinerary that never descends, at *exactly* that critical share. The finance-survivor lengths 176251, 301994, 478245, 780239 are denominators of convergents and semiconvergents of $\log 2/\log 3$ because a periodic non-descending word must realize the critical share to within $2/(n \log n \log 3)$. The walk-charge program of Section 5 and the no-momentum hypothesis are two views of one boundary — the periodic side and the almost-all side of the zero-drift line — and the period bound $L \ge 780239$ is the statement that the boundary carries no short periodic word above the floor.

The floor stratifies the failure set. By backward closure, the minimum of the failure set F (if $F \ne \emptyset$) is odd with odd image, exactly as the minimum of a cycle is (Theorem 3.2(ii), and the run form of Section 3, whose first odd run has length at least two); every odd failure with even image exceeds $N_0^{4/3} = 2.5 \cdot 10^{11}$, every even failure exceeds $N_0^2 = 1.2 \cdot 10^{17}$, and every failure that is the image $\lfloor m^{3/2} \rfloor$ of an odd m exceeds $N_0^{3/2} = 6.5 \cdot 10^{12}$ ([17, Section 6]). These are the scales at which each type of failure — cycle state or divergent start — can first appear.

The floor is a testable target. Because every $n \le N_0$ reaches 1, the statistic "$J^t(n) \le N_0$ for some $t \le d$" is a finite computation on exact orbits, and [17, Section 11] reports that for random odd starts at $y = 10^{12}$ to 10^{50} the fraction still above N_0 after $d \le 40$ steps matches the odd-start fair-coin survival within 3%. This is an observation about aggregates; it proves nothing about cycles.

What none of this does: exclude a cycle, bound a basin from above, or address the parity word of a single orbit at depth L. Paper C shows that the termination problem, cycles included, is one almost-all statement about parity words at depth $\asymp \log\log n$; Paper B shows that fixed-depth parity control, which its methods deliver through depth four, improves the constants of that statement and cannot reach it. The cycle problem as posed in this paper — the long regime $L \approx n^{0.59}$, a per-orbit parity statement at depth L — is therefore a special case of the same frontier, seen from the periodic side, and remains open.

Lean names are in Appendix A. The computational certificates are Propositions 1.3 and 5.1.

Appendix A. Lean names

The core mathematical lemmas of Sections 2–4 are mechanized in Lean 4. The names below are the corresponding theorems in `formal/Problems/Juggler/`, imported by `Problems.JugglerPaper`. Selected finite classifications of Section 3 are `native_decide` tables (Appendix D). Theorems 4.6 and 4.8 are independently certified computations. Proposition 4.9's arithmetic is Lean; its identification with $\mathcal{E}_{\text{run}}$ is Theorem 4.8.

Text	Lean
itinerary semantics	`follows_iff_itinerary`, `image_eq_iterate`, `image_append`
Theorem 2.1	`image_monotone_of_follows`
Theorem 2.2	`power_bound_word`
Corollary 2.3	`power_bound_contracts`
Theorem 2.4	`global_defect_identity`
Theorem 2.5	`global_defect_eq_zero_iff_localsTight`, `global_defect_eq_zero_implies_monochrome`, `power_bound_eq_iff_extremal`
Theorem 2.6	`global_defect_append`
Corollary 2.7	`image_eq_start_defectRatio`
per-step slack	`one_plus_eta_lt_succ_sq`
Lemma 3.1	`odd_preimage_unique`
CycleMin	`CycleMin`
Theorem 3.2	`cycle_itinerary_formally_expanding`, `cycleMin_start_odd`, `cycleMax_start_even`, `cycleMin_not_end_odd`, `square_scale_superquadratic`, `cycleMin_to_even_superquadratic`
Lemma 3.3	`lower_growth_word`
Lemma 3.4	`oo_suffix_threshold`, `ooo_suffix_threshold`, `threshold_inherits_odd_append`, `cycle_last_even_interval`, `cycle_last_even_ne_odd_sq`, `no_cycle_odd_run_append_even`, `no_cycle_itinerary_ooe`
odd-run block	`oddEvenBlock`

Text	Lean
Lemma 3.5	`no_cycle_itinerary_oooeoe`, `no_cycle_itinerary_oooooee`
Theorem 3.6	`no_cycle_itinerary_length_le_six`
Lemma 3.7	`no_cycle_itinerary_ooooeoe`, `no_cycle_itinerary_ooooooee`
Theorem 3.8	`no_cycle_itinerary_length_le_seven`, with `no_cycle_itinerary_ooeooee`, `no_cycle_itinerary_oooeooe`
Lemma 3.9	`cycle_trailing_evens_lt`
Lemma 3.10	`lowerDenom_replicate_odd`, `odd_run_lower_growth`
Lemma 3.11	`no_follows_seven_odds_of_lt256`
Theorem 3.12	`no_cycle_itinerary_two_even_ee`, `no_cycle_itinerary_two_even_eoe`
Theorem 3.13	`no_cycleMin_gapped_three_even_ee`, `no_cycleMin_gapped_three_even_eoe`
Theorem 3.14	`no_cycle_itinerary_three_even_eee`, with `no_cycle_itinerary_ooooooeee`
Theorem 3.15	`no_cycle_itinerary_three_even_eoee`
Theorem 3.16	`no_cycle_itinerary_three_even_eooee`
Theorem 3.17	`no_cycle_itinerary_three_even_eoooee`
Theorem 3.18	`no_cycle_itinerary_three_even_eeoe`
Theorem 3.19	`no_cycle_itinerary_three_even_eoeoe`
Theorem 3.20	`no_cycle_itinerary_three_even_eooeoe`
Theorem 3.21	`no_cycle_itinerary_gapped_three_even_ee`, `no_cycle_itinerary_gapped_three_even_eoe`
Theorem 3.22	`no_cycle_itinerary_even_count_le_three`
Corollary 3.23	`cycle_itinerary_length_ge_eleven`
Lemma 3.21b	canonical run form; Theorem 3.2
Lemma 3.21a	the case split of Theorem 3.22
Lemma 4.1	`log_le_two_log_add`
Lemma 4.2	`log_step_even`, `log_step_odd`
Lemma 4.3	`cycleMin_log_envelope`
Theorem 4.4	`cycleMin_finance`
Corollary 4.4c	`cycleMin_log_envelope_inv`, `cycleMin_finance_inv_sum`
Lemma 4.4b	odd-count monotonicity; human proof, not Lean
Theorem 4.6	certified identity `cycleMin_defect_finance`, per-step losses `log_floorPower_even_ge_sub`, `log_floorPower_odd_ge_sub`, invariants `cycleMin_log_le_weight`, `cycleMin_charge_prefix` (`DefectFinance.lean`); the numeric table is verified computation
Theorem 4.7	run-type packing; human proof, not Lean
Theorem 4.8	run-type table; verified computation, not Lean
Proposition 4.9	`run_survivor_unimodular`, `run_survivor_seed_F2`, `run_survivor_seed_F3`, `three_pow_step_gt_two_pow_step`, `runSurvivors_length`
Theorem 4.10	`cycleMin_gap_transfer`; abstract length bound `cycleMin_length_of_gap` (`GapTransfer.lean`)

Text	Lean
Corollary 4.11	`cycleMin_length_of_gap` with Rhin's measure [15] as hypothesis; the transcendence input is classical, not Lean
Proposition 5.1	laboratory floor; certified computation, not Lean
Theorem 5.2	raised cutoff; verified computation, not Lean
Theorem 5.3	transport inequality `cycleMin_transport`, per-step losses `log_floorPower_even_ge`, `log_floorPower_odd_ge` (`WalkTransport.lean`); §5.2 consequence `cycleMin_defect_le_charge`, `cycleMin_defect_le_hug_charge` (`WalkChargeMax.lean`)
Theorem 5.4	combinatorial core `hugOdds_le_of_admissible`; cycle-itinerary domination `cycleMin_prefix_odds_ge_hug`, `cycleMin_odds_ge_hug`; charge maximisation `stateCharge_antitone`, `hug_charge_maximal` (`WalkChargeMax.lean`); strict within-(L, o) uniqueness human
Proposition 5.5	ergodic identification human; Laplace bound Lean: `inv_sq_le_quad`, `rotation_average_le`, `rotation_average_lt`, `rotationAverage_le`, `rotationAverage_lt`, `rotationAverage_gap` (`RotationAverage.lean`)
Lemma 5.6	`budgetedWord_eq_hugWord`, `hugOdds_pow_ge`, `hugOdds_pow_lt`, `hugOdds_pow_gt`, `hugOdds_least`
Theorem 5.7	Denjoy–Koksma's variation inequality (known), not Lean; quotient arithmetic `theta_sandwich_upper`, `theta_sandwich_lower`, `lower_lt_walkTheta`, `walkTheta_lt_upper`, `cf_lower_prefix`, `cf_upper_prefix`, `theta_convergent_denominators`; DK hypotheses `theta_convergent_numerators`, `theta_convergents_unimodular`, `theta_convergents_coprime`, `theta_convergent_quality` ($\lvert\theta - p/q\rvert < 1/q^2$), `theta_block_permutations`
Theorem 5.8	digit cap Lean: general numeration `ostroDigit_le`, `ostro_sum_eq`, `ostro_digitSum_le`, instance `theta_digitSum_le`, `greedyDigitSum_le`; scan `window_digit_scan`, `window_digit_cap`, `window_digit_max`; Denjoy–Koksma comparison human
Theorem 5.9	kill template `cycleMin_hug_kill_criterion` (`DefectFinance.lean`); the per-length kill table is verified computation
Corollary 5.10	second floor and kill table; verified computation, not Lean
Corollary 5.11	third floor and kill table; verified computation, not Lean
short certificates (Section 6)	`even_finiteProgress`, `odd_even_finiteProgress`
no certificate ⇒ odd-to-odd	`no_finiteProgress_implies_odd_odd`
induction to 1	`reachesOne_of_all_finiteProgress`

Text	Lean
four-block chain	`four_block_pe_1999`

Appendix B. Admissible lengths

The record lengths of $\gamma(L)$ through 10^5, with the parity 6/5 bound $n_{\max}$ of Section 4, are

L	$o_{\min}$	$n_{\max}$
1	1	3
3	2	7
11	7	25
19	12	133
84	53	2323
569	359	23568
1054	665	788014
25781	16266	26254995
50508	31867	162848325

At the verified descent floor $N_0 = 10^6$ the first seven rows are excluded. The set $\mathcal{E} = \mathcal{E}(10^6)$ is defined by Proposition 4.4a: for each $1 \le L \le 10^5$, compute $o_{\min}(L)$ by exact integer arithmetic and $n_{\max}(L)$ from the parity inequality, and retain L if and only if $n_{\max}(L) > 10^6$. This produces 141 lengths. The first few are

$$25781,\ 26835,\ 27889,\ 28943,\ 29997,$$

and a typical combination is $26835 = 25781 + 1054$. The last is 99561. The complete list is the `lengths` array of `data/research/juggler/cycle_finance/exceptions_parity.json`. The SHA-256 of that array, serialized as a JSON list of integers with no spaces, is `dd71aa1527656ba51cb031bafa5497f7bfdbbc43151ffba2c595793326bf7944`. The SHA-256 of the whole file `exceptions_parity.json` is `6b4eec79295b70cdeb9f7db677b7fd57bcb9bd1b51177e1e74aad7b` The SHA-256 of the first-passage file `floor.json` in the same directory is `5b1ce1eec61301cf5b4f969cd5b58954255194e5c7b2` The parity table is written by `research.juggler_sequence.cycle_finance.write_parity_artifacts`. The first-passage file is regenerated by `python -m research.juggler_sequence.cycle_finance`.

The run-type table of Theorem 4.8 is `data/research/juggler/cycle_finance/budget_opt.json`. The 42 excluded lengths are `killed_by_budget`; their SHA-256, serialized as a JSON list of integers with no spaces, is `9d108776d6dc5dc1ae2594058850463cd2d3995cc57b9811471f08e3b818b90a`. The complementary 99 lengths are $\mathcal{E}_{\mathrm{run}}$. Their SHA-256, in the same serialization, is `9e2098923ccb39933630b116133a3fc2ddaf98ace4eb76dbab9b5a` The first few survivors remain

$$25781,\ 26835,\ 27889,\ 28943,\ 29997;$$

the last survivor is 99477. The three families and the unimodular basis are Proposition 4.9. The finite leftover tables of Section 3 are the Lean `native_decide` evaluations named in Appendix A (`LeftoverEval.lean`, `LeftoverShort.lean`, `LeftoverFamilies.lean`). The lattice arithmetic of Proposition 4.9 is `RunSurvivorLattice.lean`.

The laboratory instance of Section 5 has its own artifacts. The first-passage certificate of Proposition 5.1 is `data/research/juggler/cycle_finance/floor_verify/N26254995/certificate.json`; the SHA-256 of its chunk records is `cbcbb540dd860b775d2a3b4351f7cf779609104d3aaf30c342aed6b87f36c9dc`. The parity survivor list of Theorem 5.2 has SHA-256 `d5946efdfac95c715ea46c81979a0eeaf40ad8a1dc893d161bab84ffa7f2afd9`. The walk-charge kill table of Theorem 5.9 is `data/research/juggler/cycle_walk_charge/survey.json`; the SHA-256 of the walk-alive list is `225d76ad12802a690934d01e2d37b3418865441a6825a015dd883c989d8942ec`. The second-floor certificate of Corollary 5.10 is `data/research/juggler/cycle_finance/floor_verify/N162849448/certi` the SHA-256 of its chunk records is `35d9755ce52a225a161b009d0f3674b24a5d0add16035f09ea83cf3880414ae4`. Its parity survivor scan is `data/research/juggler/cycle_walk_charge/new_floor_parity_leftovers.json`, and the 15 kill records under `data/research/juggler/cycle_walk_charge/new_floor_kills/` have SHA-256 `148180cbbfba93985b7a1be455fee16db97816f1534b1feca5a4740d475aeda0` (concatenated in length order); the direct non-kill record for the survivor 478245 (margin 0.4334) is stored alongside. The third-floor certificate of Corollary 5.11 is `data/research/juggler/cycle_finance/floor_verify/N350000000/certificate.json`; the SHA-256 of its chunk records is `c57f5ccc9bce980f478dafec00c449050a5e217a4f0f3e100916c41dacbe7472`. Its parity survivor scan is `data/research/juggler/cycle_walk_charge/N350000000_parity_leftovers.json`, and the 10 kill records under `data/research/juggler/cycle_walk_charge/N350000000_kills/` have SHA-256 `d16ccfed52757d4a44368a6549a8149ccbc926472737276c577912346db854ab` (concatenated in length order); the direct non-kill record for the survivor 780239 (margin 0.6049) is stored alongside. The exact-integer CPU computation with guarded comparisons is the authoritative record for the first- and second-floor claims in this note. The third-floor kill table is the GPU floating-point certified comparison (`research.juggler_sequence.cycle_walk_charge_gpu`), the same IEEE-double kernel previously checked against stored CPU records to relative error below 10^{-13}; its performance figures live in the repository documentation. The probes are `research.juggler_sequence.cycle_walk_charge` (transport and kill table), `cycle_walk_ostrowski` (certified quotient sandwich and block envelope), and `cycle_walk_window` (window scan); artifacts live under `data/research/juggler/cycle_walk_*`.

Reproducibility of the second-floor kill table. From a fresh checkout of the repository, `pip install -e .` followed by, for a survivor length L,

```
python -c "from research.juggler_sequence.cycle_walk_charge \
import certified_report; print(certified_report(L, 162849448))"
```

recomputes the authoritative certified record behind Corollary 5.10 for that length (at $L = 478245$ it prints the non-kill); the same call through the GPU port, `python -m research.juggler_sequence.cycle_walk_charge_gpu L 162849448`, reproduces it in seconds per length on one consumer GPU. The first-floor kill table of Theorem 5.9 is `python -m research.juggler_sequence.cycle_walk_charge --survey`.

Reproducibility of the third-floor kill table. The committed records behind Corollary 5.11 are the GPU certified reports. From a fresh checkout,

```
python -c "from research.juggler_sequence.cycle_walk_charge_gpu \
import gpu_certified_report; print(gpu_certified_report(L, 350000000))"
```

recomputes the record for length L at this floor (at $L = 780239$ it prints the non-kill). Do not pass `--write` to the module CLI: that path is hardcoded to the second-floor directory.

Appendix C. Exact floor defect

This appendix records the exact identity behind Theorem 2.2. The present note uses only the nonnegativity $\Delta_w(n) \geq 0$. A nontrivial itinerary-dependent lower bound on Δ_w is left for future work.

For a single branch,

$$x^e = J(x)^2 + \rho(x), \qquad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \leq \rho(x) < 2J(x) + 1$. Write $\operatorname{gap}(a, \rho, e) = (a + \rho)^e - a^e$. The *global defect* $\Delta_w(n)$ is the terminal value of the resulting power-gap recurrence.

Theorem 2.4 (global defect identity). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \qquad \Delta_w(n) \geq 0.$$

Theorem 2.2 is the inequality $\Delta_w(n) \geq 0$.

Proof. Induct on w. The empty itinerary is $n = n + 0$. An even letter substitutes $x = J(x)^2 + \rho(x)$ into the inductive identity and lifts the new remainder through 2^ℓ. An odd letter cubes the identity and then substitutes $x^3 = J(x)^2 + \rho(x)$. Each step adds a nonnegative power-gap. □

For a one-letter illustration, $n = 3$ and $w = O$ give $J(3) = 5$ and $3^3 = 27 = 5^2 + 2$, so $\Delta_O(3) = 2$.

Theorem 2.5 (vanishing). If w is realized at n, the following are equivalent: $\Delta_w(n) = 0$; every local remainder along w vanishes; and $\left(J^{|w|}(n)\right)^{2^{|w|}} = n^{3^{\#O(w)}}$. In that case w is monochrome: either $w = E^k$ and $n = a^{2^k}$ for an even a, or $w = O^k$ and $n = a^{2^k}$ for an odd a. A realized mixed itinerary therefore has $\Delta_w(n) > 0$.

Proof. Theorem 2.4 gives the first and third items. A power-gap vanishes if and only if its addend vanishes, so zero defect forces every remainder to vanish, and conversely. Vanishing remainders preserve parity, hence monochrome itineraries, and unique factorization produces the two power towers. □

Theorem 2.6 (composition). If u is realized at n and v is realized at $m = J^{|u|}(n)$, then $\Delta_{uv}(n)$ is the sum of two power-gaps, one lifting $\Delta_u(n)$ through $3^{\#O(v)}$ and one lifting $\Delta_v(m)$ through $2^{|u|}$. In particular $\Delta_v(m)^{2^{|u|}} \le \Delta_{uv}(n)$, and every local remainder satisfies $\rho_j^{2^j} \le \Delta_w(n)$.

Proof. Apply Theorem 2.4 to u, to v, and to uv, and expand the two power-gaps. □

Composition is polynomial, not additive. The Lean form is `global_defect_append`.

Corollary 2.7 (cycle surplus). If w is a cycle itinerary at n, then $\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$ exactly.

Proof. Theorem 2.4 with $m = n$. □

A mixed itinerary has strict total defect, but the relative slack of a single letter tends to 0 with the state, so no uniform local tax exists. Theorem 4.4 does not use these identities.

Appendix D. Family exclusions

This appendix records the small-cycle censuses (Lemma 3.5, Theorem 3.6, Lemma 3.7, Theorem 3.8) and the family-by-family exhaustion used by Lemma 3.21a and Theorem 3.22. Each geometry is killed by a next-square obstruction, by long odd-run growth against a last-even one-step preimage, or by a finite exceptional window.

Lemma 3.5 (two length-six exclusions). Neither *OOOEOE* nor *OOOOEE* is a cycle itinerary at any $n \ge 2$.

Proof. First, if $n \ge 256$, then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed $257^{64} < 2{\cdot}256^{64}$ because $(1{+}1/256)^{64} < e^{64/256} = e^{1/4} < 2$, and $256(n{+}1) \le 257\,n$, so $(n{+}1)^{64} < 2n^{64}$. Then $2^{130}(n{+}1)^{64} < 2^{131}n^{64}$. Since $n \ge 256 = 2^8$, one has $n^{17} \ge 2^{136}$, and therefore $2^{131}n^{64} < 2^{136}n^{64} \le n^{81}$.

For $2 \le n < 256$, neither word returns to its start. This is a table of 254 evaluations of each word: at every start, either some letter fails to match the current parity, or the six-step image differs from the start. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_itinerary_oooeoe` and `no_cycle_itinerary_ooooee` (Appendix A).

Now suppose *OOOOEE* is a cycle itinerary at $n \ge 256$, and write $z = J^4(n)$ for the image after the prefix *OOOO*. Lemma 3.4(iv) gives $J(z) < (n+1)^2$, and the preceding even letter gives $z < (J(z)+1)^2$, hence $z < (n+1)^4$. Lemma 3.3 on *OOOO* gives $n^{81} \le 2^{130}z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose *OOOEOE* is a cycle itinerary at $n \ge 256$. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor\sqrt{z_3}\rfloor$, so $z_3 < (y+1)^2$. Lemma 3.3 on *OOO* gives $n^{27} \le 2^{38}z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters *OE* give the odd-preimage bound $y^3 < (n+1)^4$. Write $A = n+1 \ge 257$. We claim $(y+1)^3 < 2A^4$. If $y \le A$, this is $(A+1)^3 < 2A^4$. If $y > A$, then $y \ge 258$, so $3y+1 < y^2$ and hence $(y+1)^3 = y^3+3y^2+3y+1 < A^4+4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \le yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. □

Theorem 3.6 (small-cycle census). No itinerary of length at most six is a cycle itinerary at any $n \ge 2$.

Proof. Rotating a cycle itinerary by one letter moves the base point one step along the trajectory and yields another cycle itinerary; every state of the cycle is at least 2, because a trajectory that reaches 1 stays at 1 and cannot return to a start $n \geq 2$.

If every letter is odd, the start is odd, hence $n \geq 3$, and the odd branch strictly increases there: $J(x) > x$ for odd $x \geq 3$, since $x^3 \geq (x+1)^2$. The trajectory ascends strictly and never returns. Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle itinerary vE of the same length based at a cycle state $m \geq 2$. It therefore suffices to exclude even-terminating cycle itineraries of length at most six.

By Theorem 3.2(i) a cycle itinerary is formally expanding. No even-terminating itinerary of length one or two is expanding ($2 > 3^0$ and $4 > 3$).

Length three: the only expanding candidate is OOE. If $m \geq 5$ realizes OO, Lemma 3.4(i) gives $J^2(m) \geq (m+1)^2$, contradicting Lemma 3.4(iv). The only smaller odd start is $m = 3$, where $J^2(3) = 11$ is odd, so the final even letter is not realized.

Length four: the expanding filter requires three odd letters among the first three, leaving only O^3E, excluded by Lemma 3.4(v).

Length five: the filter requires four odd letters among the first four, leaving only O^4E, excluded by Lemma 3.4(v).

Length six: the filter requires at least four odd letters among the first five, leaving O^5E, $EOOOOE$, $OEOOOE$, $OOEOOE$, $OOOEOE$, and $OOOOEE$. Lemma 3.4(v) excludes O^5E. The itinerary $EOOOOE$ rotates one step onto $OOOOEE$, and $OEOOOE$ rotates two steps onto $OOOEOE$; both are excluded by Lemma 3.5.

It remains to exclude $OOEOOE$. Let this itinerary be a cycle itinerary at some start, and rotate to a cycle minimum $m \geq 2$. The three alignments of the two even letters are $OOEOOE$, $OEOOEO$, and $EOOEOO$. The last starts even, so it cannot be a minimum, by Theorem 3.2(ii). The middle starts OE: the first image is even and strictly below m^2, whereas every even state after a cycle minimum is at least m^2, as proved in Theorem 3.2(iii). Thus the minimum orientation is $OOEOOE$. In particular m is odd, so $m \geq 3$. The prefix OOE must be realized. If $m = 3$, then $3 \to 5 \to 11$ and 11 is odd, so OOE is not realized. Hence $m \geq 5$. Write $y = J^3(m)$ for the state after OOE. Then $y \geq m$ by minimality, so $y \geq 5$. The suffix OO is realized at y, and Lemma 3.4(i) gives $J^2(y) \geq (y+1)^2 \geq (m+1)^2$. The last letter is even, so Lemma 3.4(iv) gives $J^2(y) < (m+1)^2$. □

Lemma 3.7 (two length-seven exclusions). Neither $OOOOEOE$ nor $OOOOOEE$ is a cycle itinerary at any $n \geq 2$.

Proof. First, if $n \geq 14$, then

$$n^{243} > 2^{422}(n+1)^{128}.$$

Indeed $14(n+1) \leq 15n$, so $(n+1)^{128} \leq (15/14)^{128} n^{128}$. The finite comparison $2^{422}15^{128} < 14^{243}$ then yields $2^{422}(n+1)^{128} < 14^{115} n^{128} \leq n^{243}$.

For $2 \leq n < 14$, neither word is realized: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_itinerary_ooooeoe` and `no_cycle_itinerary_oooooee` (Appendix A).

Now suppose $OOOOOEE$ is a cycle itinerary at $n \geq 14$, and write $z = J^5(n)$ for the image after the prefix $OOOOO$. Lemma 3.4(iv) on the last even letter, together with the preceding even letter, gives $z < (n+1)^4$. Lemma 3.3 on $OOOOO$ gives $n^{243} \leq 2^{422} z^{32}$, so $n^{243} < 2^{422}(n+1)^{128}$, contradicting the tail inequality.

Finally suppose $OOOOEOE$ is a cycle itinerary at $n \geq 14$. Write $z_4 = J^4(n)$ and $y = J(z_4) = \lfloor \sqrt{z_4} \rfloor$, so $z_4 < (y+1)^2$. Lemma 3.3 on $OOOO$ gives $n^{81} \leq 2^{130} z_4^{16} < 2^{130}(y+1)^{32}$. Cubing yields $n^{243} < 2^{390}(y+1)^{96}$. The last letters OE give the odd-preimage bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 15$. The same comparison $(y+1)^3 < 2A^4$ as in Lemma 3.5 holds at this smaller scale. Raising to the thirty-second power gives $(y+1)^{96} < 2^{32}(n+1)^{128}$. Combining with the cubed lower envelope produces again $n^{243} < 2^{422}(n+1)^{128}$. □

Theorem 3.8 (small-cycle census through length seven). No itinerary of length at most seven is a cycle itinerary at any $n \geq 2$.

Proof. The reduction of Theorem 3.6 applies at every length: an all-odd itinerary cannot return, and every mixed cycle itinerary rotates to an even-terminating cycle itinerary based at a cycle state $m \geq 2$. Lengths at most six are Theorem 3.6. It remains to exclude even-terminating cycle itineraries of length seven.

By Theorem 3.2(i) a cycle itinerary is formally expanding. The even-terminating expanding length-seven words are exactly O^6E, EO^5E, OEO^4E, $OOEO^3E$, O^3EO^2E, O^4EOE, and O^5EE. Lemma 3.4(v) excludes O^6E. The itinerary EO^5E rotates one step onto O^5EE, and OEO^4E starts OE, so it cannot be a cycle minimum (Theorem 3.2(iii)) and rotates two steps onto O^4EOE; both leftovers are excluded by Lemma 3.7.

It remains to exclude $OOEO^3E$ and O^3EO^2E. Rotate either word to a cycle minimum $m \geq 2$. For $OOEO^3E$ the minimum orientation retains the internal even letter followed by the suffix OOO. Then $m \geq 3$, the prefix through that even letter is realized, and Lemma 3.4(ii) at threshold 3 contradicts the last even one-step preimage. For O^3EO^2E the same bootstrap applies with suffix OO and threshold 5, once $m = 3$ is removed: at $m = 3$ the state after OOO is even, so the next even letter is not realized. □

Theorem 3.12 (two-even leftover families). Let $k \geq 6$ and $n \geq 2$. Neither $O^{k-2}EE$ nor $O^{k-3}EOE$ is a cycle itinerary at n.

Proof. First, if $n \geq 256$, then

$$n^{3^{k-2}} > 2^{e_{k-2}}(n+1)^{2^k}.$$

The case $k = 6$ is the tail inequality of Lemma 3.5. If the display holds at some $k \geq 6$, cubing both sides produces

$$n^{3^{k-1}} > 2^{3e_{k-2}}(n+1)^{3\cdot 2^k}.$$

The recurrence of Lemma 3.10 gives $e_{k-1} = 3e_{k-2} + 2^{k-1}$, so the desired comparison at length $k+1$ reduces to $2^{2^{k-1}} < (n+1)^{2^k}$. Equivalently $2 < (n+1)^2$, which holds for every $n \geq 256$.

Now suppose $O^{k-2}EE$ is a cycle itinerary at such an n. Write $z = J^{k-2}(n)$. Lemma 3.9 with $r = 2$ gives $z < (n+1)^4$. Lemma 3.10 on the prefix O^{k-2} gives $n^{3^{k-2}} \leq 2^{e_{k-2}}z^{2^{k-2}}$, hence $n^{3^{k-2}} < 2^{e_{k-2}}(n+1)^{2^k}$, contradicting the tail.

Finally suppose $O^{k-3}EOE$ is a cycle itinerary at such an n. Write $z = J^{k-3}(n)$ and $y = \lfloor\sqrt{z}\rfloor$, so $z < (y+1)^2$. Lemma 3.10 on O^{k-3} and cubing produce $n^{3^{k-2}} < 2^{3e_{k-3}}(y+1)^{3\cdot 2^{k-2}}$. The last letters OE give the odd-preimage bound $y^3 < (n+1)^4$. The comparison $(y+1)^3 < 2(n+1)^4$ of Lemma 3.5 applies at this scale. Raising it to the power 2^{k-2} and using $e_{k-2} = 3e_{k-3} + 2^{k-2}$ recovers again $n^{3^{k-2}} < 2^{e_{k-2}}(n+1)^{2^k}$.

For $2 \leq n < 256$, the cases $k = 6$ and $k = 7$ are Lemmas 3.5 and 3.7. The remaining short words O^6EE, O^5EOE, and O^6EOE fail to return on the same 254-start window; this is the Lean `native_decide` evaluation behind `no_cycle_itinerary_two_even_ee` and `no_cycle_itinerary_two_even_eoe` (Appendix A). Any longer leftover of either family begins with seven consecutive odd letters, which Lemma 3.11 forbids on this window. □

Theorem 3.13 (first-even transport). Let $n \geq 2$. No minimum-based cycle itinerary at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Write $y = J^{a+1}(n)$ for the state after the first even letter. Minimum-basedness gives $y \geq n$. In the first family the remainder after that letter is O^bEE with $b+2 \geq 6$; in the second it is O^bEOE with $b+3 \geq 6$. The trailing-even and last-odd one-step preimages of those remainders are measured against the cycle start n. Combined with Lemma 3.10 at the remainder start y, the same algebra as in Theorem 3.12 produces

$$y^{3^{\ell-2}} < 2^{e_{\ell-2}}(n+1)^{2^\ell} \leq 2^{e_{\ell-2}}(y+1)^{2^\ell},$$

where ℓ is the remainder length. If $y \geq 256$, the first paragraph of Theorem 3.12 supplies the opposite inequality at y.

If $y < 256$, then $n \leq y < 256$. A gapped leftover of total length at least 17 has $a \geq 7$ or $b \geq 7$, so either the prefix or the remainder realizes seven consecutive odd letters, contradicting Lemma 3.11. The finitely many short-gap words with $2 \le a \le 6$ and $b \le 6$ fail to be minimum-based cycle itineraries on the window $2 \le n < 256$; this is the Lean `native_decide` evaluation behind `no_cycleMin_gapped_three_even_ee` and `no_cycleMin_gapped_three_even_eoe` (Appendix A). □

Theorem 3.14 (three trailing evens). Let $a \ge 6$ and $n \ge 2$. The itinerary O^aEEE is not a cycle word at n.

Proof. Write $z = J^a(n)$. Lemma 3.9 with $r = 3$ gives $z < (n+1)^8$. Lemma 3.10 then yields $n^{3^a} < 2^{e_a}(n+1)^{2^{a+3}}$ on any such cycle itinerary. For $n \ge 128$ the opposite comparison

$$n^{3^a} > 2^{e_a}(n+1)^{2^{a+3}}$$

holds. The case $a = 6$ is $n^{729} > 2^{1330}(n+1)^{512}$. Indeed, for $n \ge 128$ one has $(n+1)^{512} < (129/128)^{512}n^{512} < e^4n^{512} < 64\,n^{512}$, so the claimed bound reduces to $n^{217} > 2^{1336}$. Since $n \ge 128 = 2^7$, the left side is at least 2^{1519}. Cubing the $a = 6$ comparison produces the general case: the recurrence of Lemma 3.10 reduces the inductive step to $(n+1)^4 > 2$, which holds for every $n \ge 128$.

For $2 \le n < 128$, the case $a = 6$ is a table of evaluations of *OOOOOOEEE*: at every such start the itinerary fails to return. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_itinerary_ooooooeee` (Appendix A). For $a \ge 7$ the prefix contains seven consecutive odd letters, which Lemma 3.11 forbids on this window. □

Theorem 3.15 (mixed bunched family *EOEE*). Let $a \ge 5$ and $n \ge 2$. The itinerary O^aEOEE is not a cycle word at n.

Proof. First let $n \ge 4$, and write $z = J^a(n)$, $y = \lfloor\sqrt{z}\rfloor$, and $p = J(y)$. Lemma 3.9 with $r = 2$ after the prefix O^aEO gives $p < (n+1)^4$. The letter after y is odd, so Lemma 3.10 at length one yields $y^3 \le 4p^2 < 4(n+1)^8$. For $n \ge 4$ one has $4(n+1)^8 < (n+1)^9$, hence $y < (n+1)^3$. The even one-step preimage at z then gives $z < (y+1)^2 \le (n+1)^6$. Combined with Lemma 3.10, any such cycle itinerary would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{6\cdot 2^a}.$$

For $a = 5$ and $n \ge 314$, the opposite comparison $n^{243} > 2^{422}(n+1)^{192}$ holds. The base instance is the finite inequality $2^{422}315^{192} < 314^{243}$. If the display holds at some $n \ge 1$, the elementary comparison $n(n+2) < (n+1)^2$ upgrades it to the same display at $n+1$. Cubing then produces the comparison at $a+1$, once $(n+1)^6 > 4$. In particular the case $a = 6$ already holds for every $n \ge 16$.

For $2 \le n < 314$ and $a = 5$, and for $2 \le n < 16$ and $a = 6$, the itinerary fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_itinerary_three_even_eoee` (Appendix A). For $a \ge 7$ and $n < 256$, Lemma 3.11 applies. For $a \ge 6$ and $n \ge 16$, the tail of the previous paragraph applies. □

Theorem 3.16 (mixed bunched family *EOOEE*). Let $a \ge 4$ and $n \ge 2$. The itinerary O^aEOOEE is not a cycle itinerary at n.

Proof. First let $n \ge 32$, and write $z = J^a(n)$, $y = \lfloor\sqrt{z}\rfloor$, and p for the image after the prefix O^aEOO. Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The two letters after y are odd, so Lemma 3.10 at length two yields $y^9 \le 2^{10}p^4 < 2^{10}(n+1)^{16}$. For $n \ge 32$ one has $2^{10} < (n+1)^2$, hence $y < (n+1)^2$. The even one-step preimage at z then gives $z < (y+1)^2 \le (n+1)^4$. Combined with Lemma 3.10, any such cycle itinerary would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \ge 256$ and $a \ge 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a+2$.

For $2 \le n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_itinerary_three_even_eooee` (Appendix A). For $a \ge 7$, Lemma 3.11 applies. □

Theorem 3.17 (mixed bunched family *EOOOEE*). Let $a \ge 3$ and $n \ge 2$. The itinerary $O^aEOOOEE$ is not a cycle itinerary at n.

Proof. First let $a \geq 4$ and $n \geq 3$, and write $z = J^a(n)$, $y = \lfloor\sqrt{z}\rfloor$, and p for the image after the prefix O^aEOOO. Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The three letters after y are odd, so Lemma 3.10 at length three yields $y^{27} \leq 2^{38}p^8 < 2^{38}(n+1)^{32}$. For $n \geq 3$ one has $2^{38} < (n+1)^{22}$, hence $y < (n+1)^2$. The even one-step preimage at z then gives $z < (y+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle itinerary would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a+2$, already used in Theorem 3.16.

Now let $a = 3$ and $n \geq 256$. The same two-even one-step preimage gives $z < (y+1)^2$, so Lemma 3.10 at the prefix O^3 yields $n^{27} < 2^{38}(y+1)^{16}$. The three-odd envelope on y still gives $y^{27} < 2^{38}(n+1)^{32}$.

If $y < 39$, then $n^{27} < 2^{38} \cdot 39^{16}$. The numerical comparison $2^{38} \cdot 39^{16} < 24^{27}$ contradicts $n \geq 24$.

If $y \geq 39$, the numerical comparison $40^{27} < 2 \cdot 39^{27}$ upgrades to $(y+1)^{27} < 2y^{27}$, hence $(y+1)^{27} < 2^{39}(n+1)^{32}$. Cubing the prefix bound three times produces $n^{729} < 2^{1026}(y+1)^{432}$. Raising the successor bound to the sixteenth power produces $(y+1)^{432} < 2^{624}(n+1)^{512}$. Combining these displays yields $n^{729} < 2^{1650}(n+1)^{512}$. The opposite comparison holds at $n = 197$ and persists to every larger start by the elementary comparison $n(n+2) < (n+1)^2$ already used in Theorem 3.15.

For $2 \leq n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_itinerary_three_even_eoooee` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. $\square$

Theorem 3.18 (mixed bunched family $EEOE$). Let $a \geq 5$ and $n \geq 2$. The itinerary O^aEEOE is not a cycle itinerary at n.

Proof. First let $n \geq 4$, and write $z = J^a(n)$ and y for the last odd letter of O^aEEOE. The suffix EOE is a cycle suffix, so the last-odd cube of Theorem 3.15 gives $y^3 < (n+1)^4$. The two letters between z and y are even, so $z < (y+1)^4$. For $n \geq 4$ the successor comparison $(y+1)^3 < 2(n+1)^4$ upgrades this to $z < (n+1)^6$. Combined with Lemma 3.10, any such cycle itinerary would satisfy the same display as Theorem 3.15:

$$n^{3^a} < 2^{e_a}(n+1)^{6 \cdot 2^a}.$$

The opposite comparison is therefore the tail of Theorem 3.15: it holds for $a = 5$ and $n \geq 314$, and already for $a = 6$ and $n \geq 16$.

For $2 \leq n < 314$ and $a = 5$, and for $2 \leq n < 16$ and $a = 6$, the itinerary fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_itinerary_three_even_eeoe` (Appendix A). For $a \geq 7$ and $n < 256$, Lemma 3.11 applies. For $a \geq 6$ and $n \geq 16$, the tail of the previous paragraph applies. $\square$

Theorem 3.19 (mixed bunched family $EOEOE$). Let $a \geq 4$ and $n \geq 2$. The itinerary O^aEOEOE is not a cycle itinerary at n.

Proof. First let $n \geq 32$, and write $z = J^a(n)$, $w = \lfloor\sqrt{z}\rfloor$, and y for the last odd letter. The suffix EOE again gives $y^3 < (n+1)^4$. The one-odd envelope on w yields $w^3 \leq 4s^2$, where s is the image after O^aEO. The last-odd one-step preimage and $n \geq 32$ upgrade this to $w < (n+1)^2$, hence $z < (w+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle itinerary would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the shared tail already used in Theorem 3.16.

For $2 \leq n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_itinerary_three_even_eoeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. $\square$

Theorem 3.20 (mixed bunched family $EOOEOE$). Let $a \geq 3$ and $n \geq 2$. The itinerary $O^aEOOEOE$ is not a cycle itinerary at n.

Proof. First let $a \geq 4$ and $n \geq 4$, and write $z = J^a(n)$, $u = \lfloor\sqrt{z}\rfloor$, and y for the last odd letter. The suffix EOE gives $y^3 < (n+1)^4$, hence $(y+1)^3 < 2(n+1)^4$. The two letters after u are odd, so Lemma 3.10 at length two yields $u^9 \leq 2^{10}s^4 < 2^{10}(y+1)^8$. Cubing that display against the last-odd successor bound produces $u^{27} < 2^{38}(n+1)^{32}$. The same comparison as in Theorem 3.17 then gives $u < (n+1)^2$, hence $z < (u+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle itinerary would satisfy the shared two-even tail of Theorem 3.16.

Now let $a = 3$ and $n \geq 256$. The prefix O^3 against $z < (u+1)^2$ yields $n^{27} < 2^{38}(u+1)^{16}$, and the two-odd plus last-odd geometry of the previous paragraph yields $u^{27} < 2^{38}(n+1)^{32}$. These are the same two displays as in the $a = 3$ case of Theorem 3.17, with u in place of y, and the same small/large split applies.

For $2 \leq n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_itinerary_three_even_eooeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. □

Theorem 3.21 (gapped leftovers as cycle itineraries). Let $n \geq 2$. No cycle itinerary at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Every cycle itinerary has a minimum-based rotation. It is therefore enough to check that every cyclic shift of either word is an already-excluded cycle-minimum orientation.

Write w for the gapped word. In the first family, $|w| = a + b + 3$. The rotation by $k = 0$ is the original itinerary, excluded as a cycle minimum by Theorem 3.13. The rotation by $k = a+1$ is the bootstrap itinerary O^bEEO^aE. That itinerary has an internal even letter and last gap at least 2. If $a \geq 3$, the last-gap threshold of Lemma 3.4 at OOO and $N = 3$ excludes it. If $a = 2$, the same lemma at OO and $N = 5$ excludes every start $n \geq 5$; the remaining odd start $n = 3$ does not realize four consecutive odd letters, so it cannot follow O^b for $b \geq 4$. The rotation by $k = a + b + 2$ begins with the last even letter of w, which Theorem 3.2 forbids at a cycle minimum. Every other rotation ends with an odd letter, likewise forbidden at a cycle minimum.

In the second family, $|w| = a + b + 4$. The same four classes appear: the original word is Theorem 3.13; the rotation by $k = a + 1$ is O^bEOEO^aE, excluded by the same last-gap thresholds, with the remaining start $n = 3$ and $a = 2$ either failing to realize four odds or, when $b = 3$, reaching 6 after $OOOE$ and then meeting an odd letter; the rotation by $k = a + b + 2$ begins OE, forbidden by Theorem 3.2; and every other rotation ends odd.

The original start need not be a cycle minimum. After rotation the start is a minimum, so the hypothesis $y < n$ that blocked Theorem 3.13 does not arise. □

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